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(54) **METHODS FOR MEDICAL IMAGE
VISUALIZATION**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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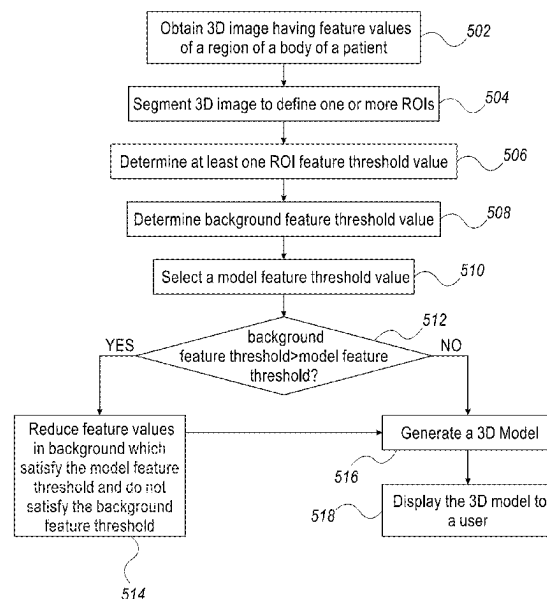
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(57) **ABSTRACT**

A computer-implemented method includes obtaining a three-dimensional (3D) image of a region of a body of a patient, the 3D image having feature values. The 3D image is segmented to define one or more regions of interest (ROIs). At least one region of interest (ROI) feature threshold is determined. A background feature threshold is determined. A 3D model is generated from the 3D image based on the determined at least one ROI feature threshold, the determined background feature threshold, and the segmentation. The 3D model is output for display to a user.

20 Claims, 7 Drawing Sheets



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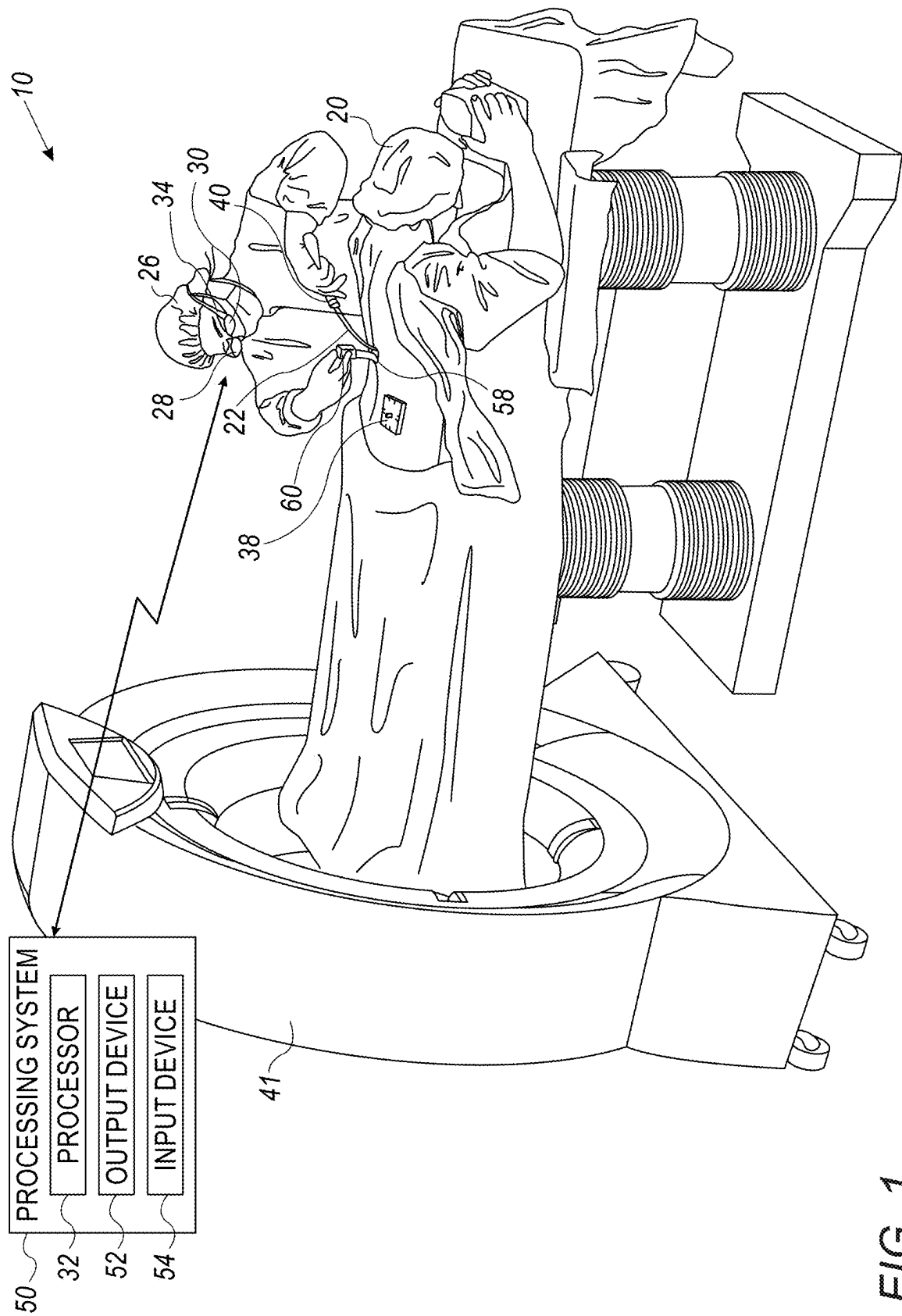
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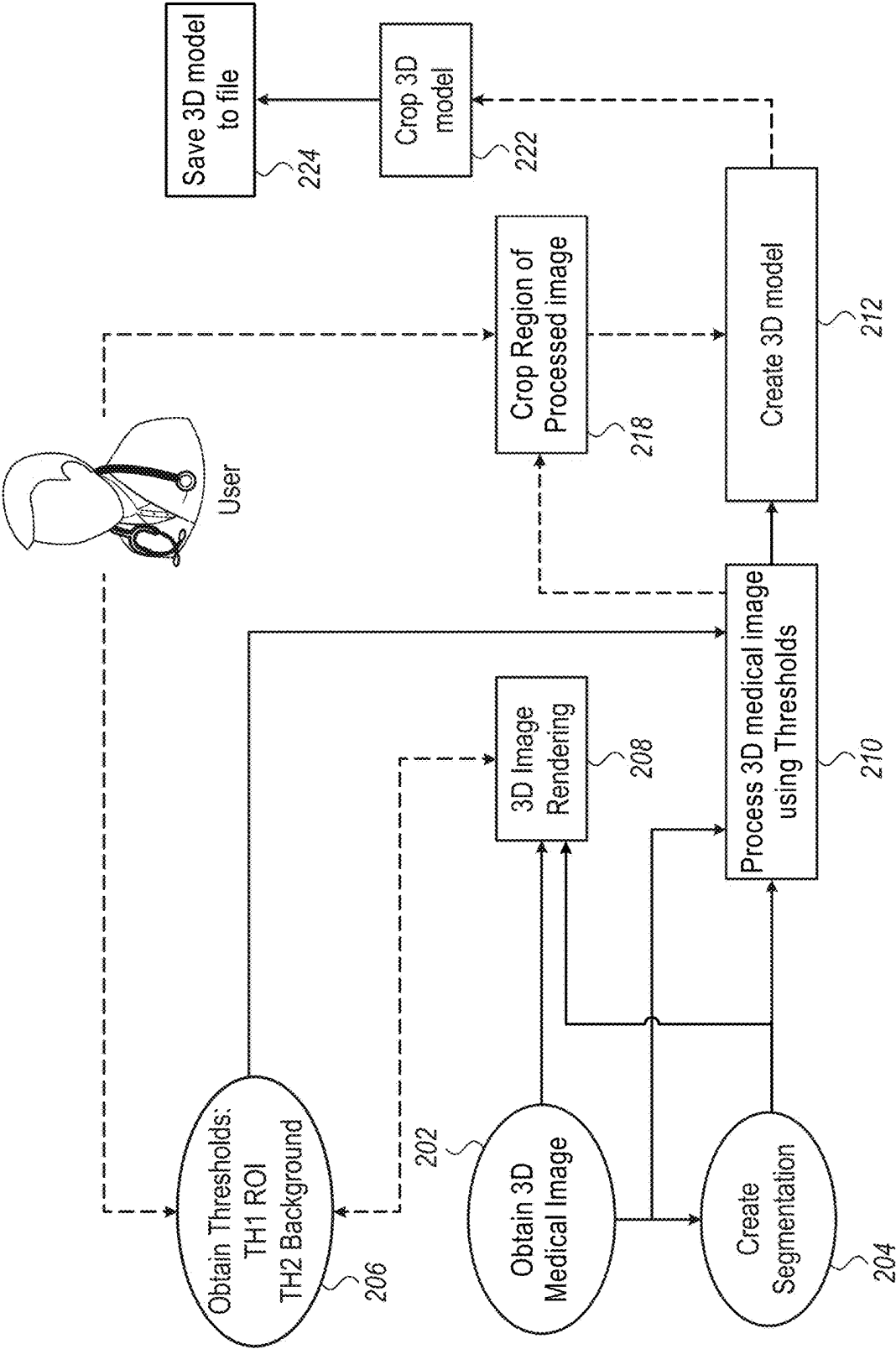


FIG. 2

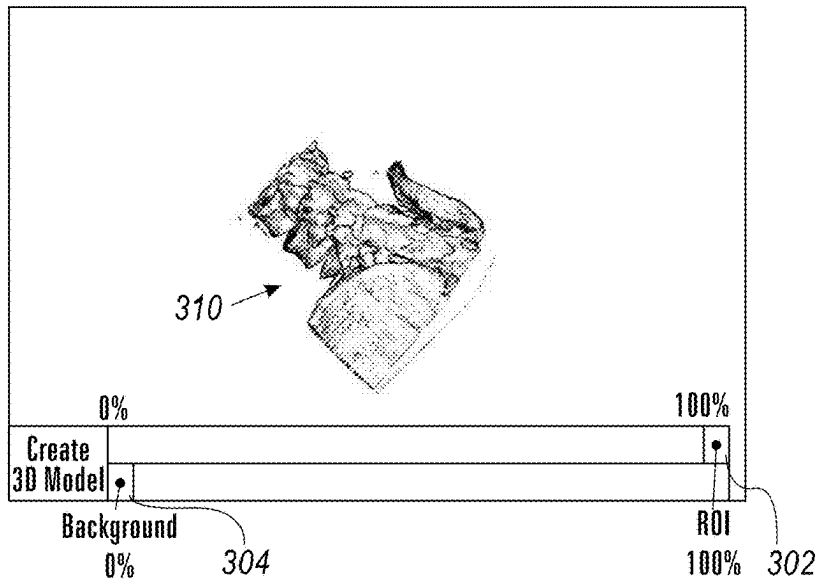


FIG. 3A

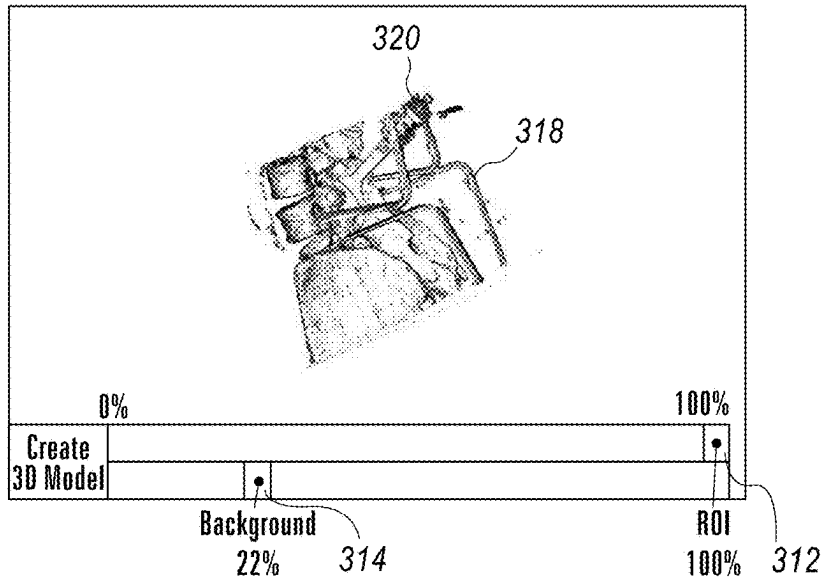


FIG. 3B

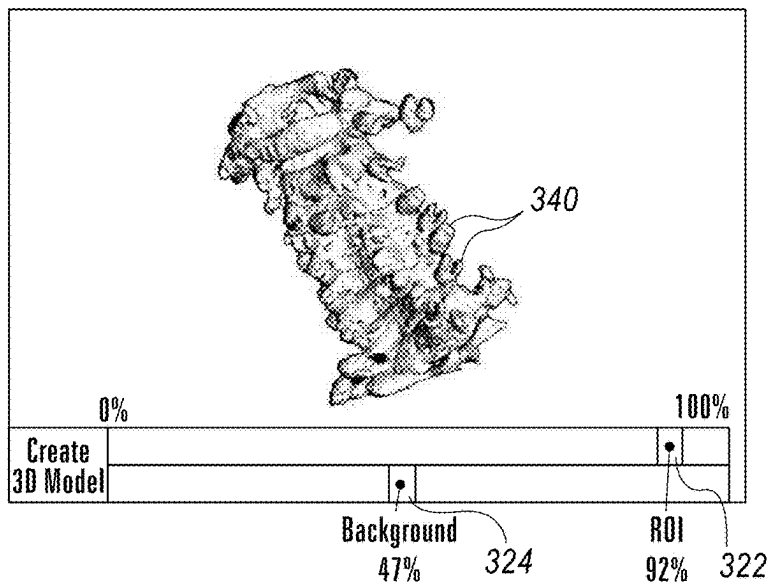
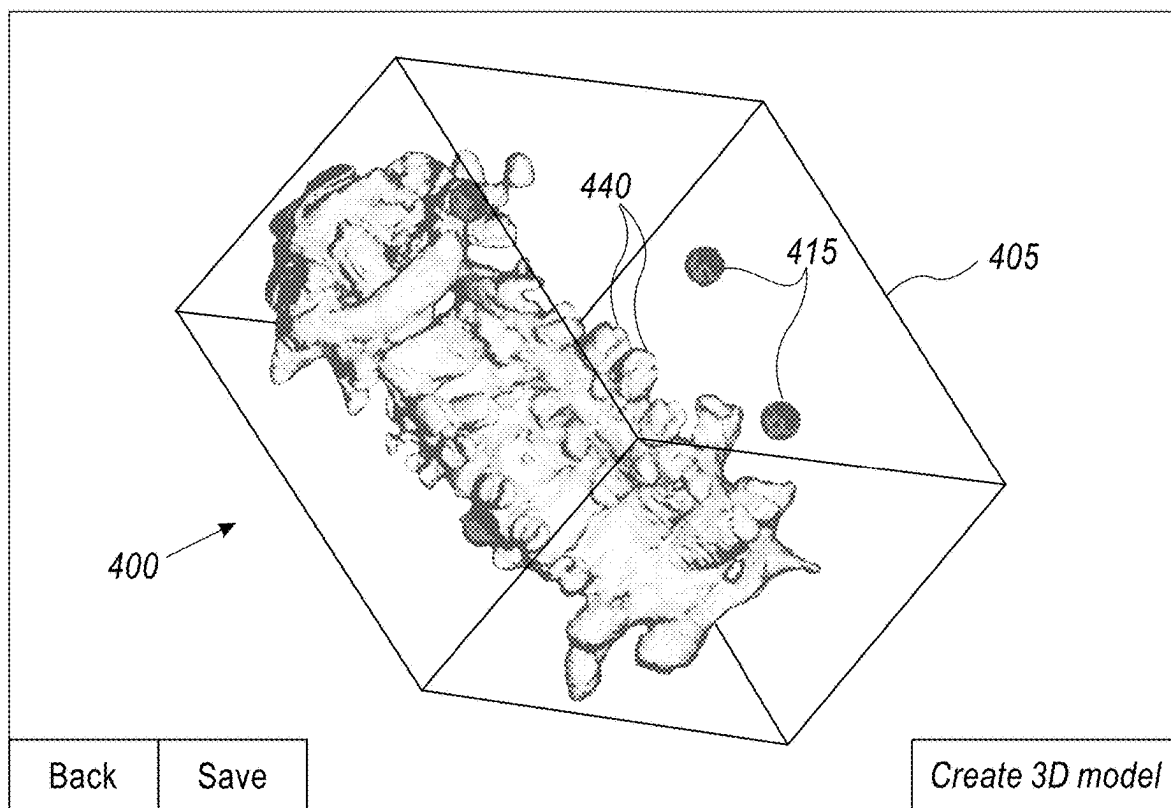


FIG. 3C

**FIG. 4**

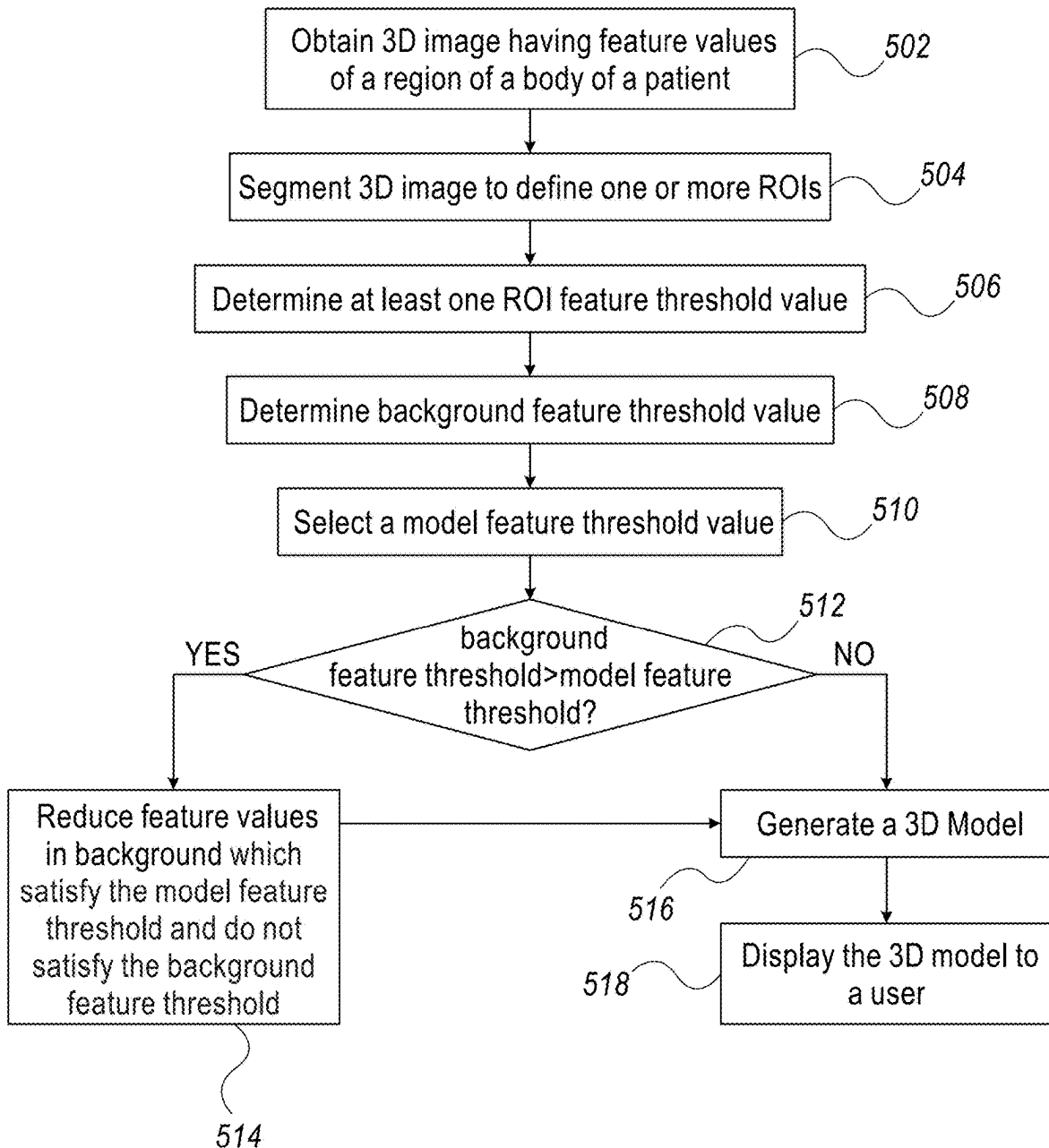


FIG. 5

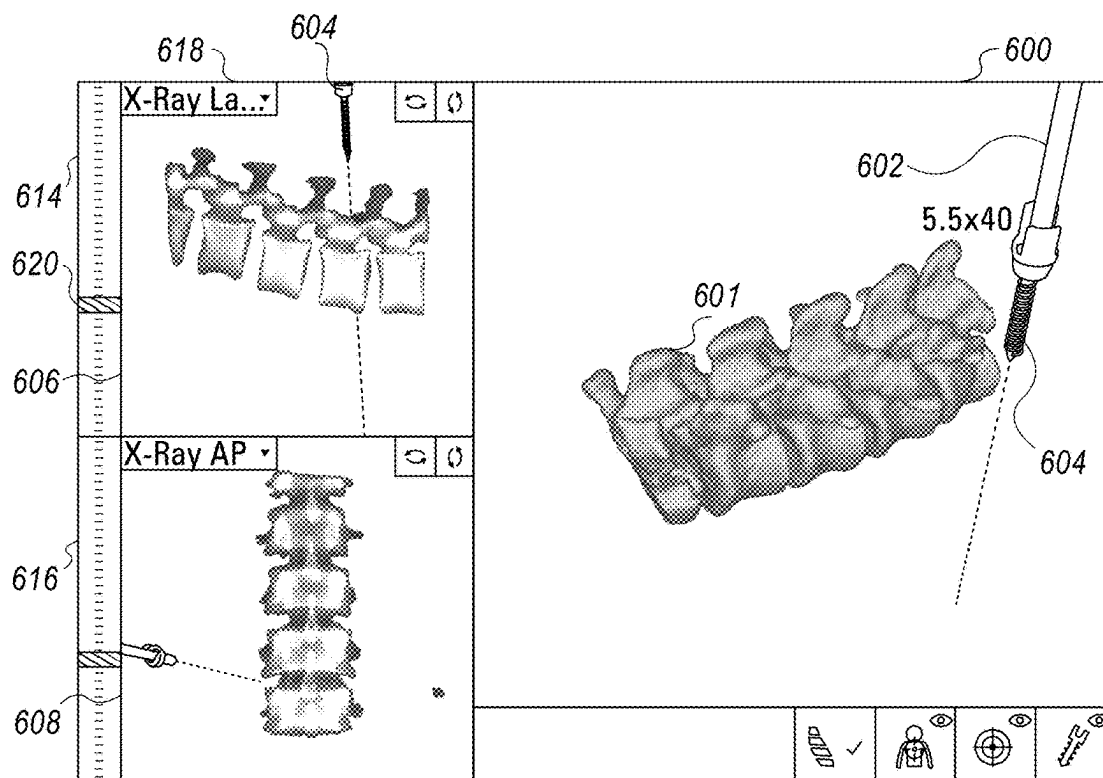


FIG. 6A

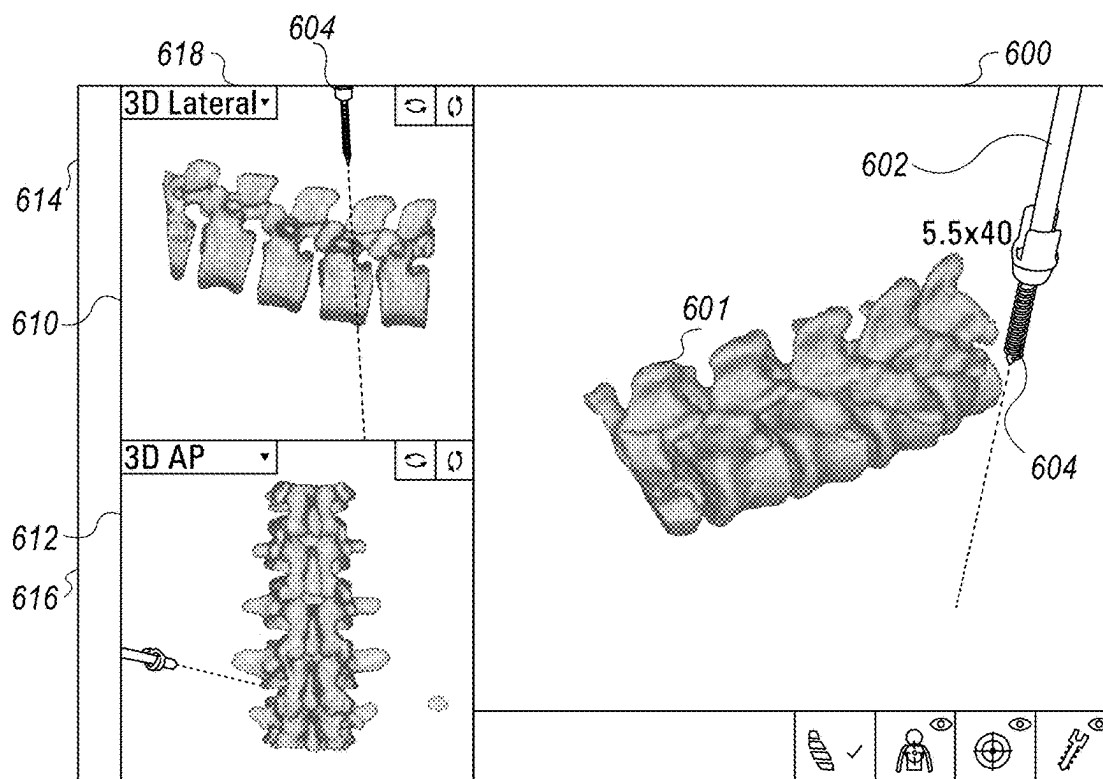


FIG. 6B

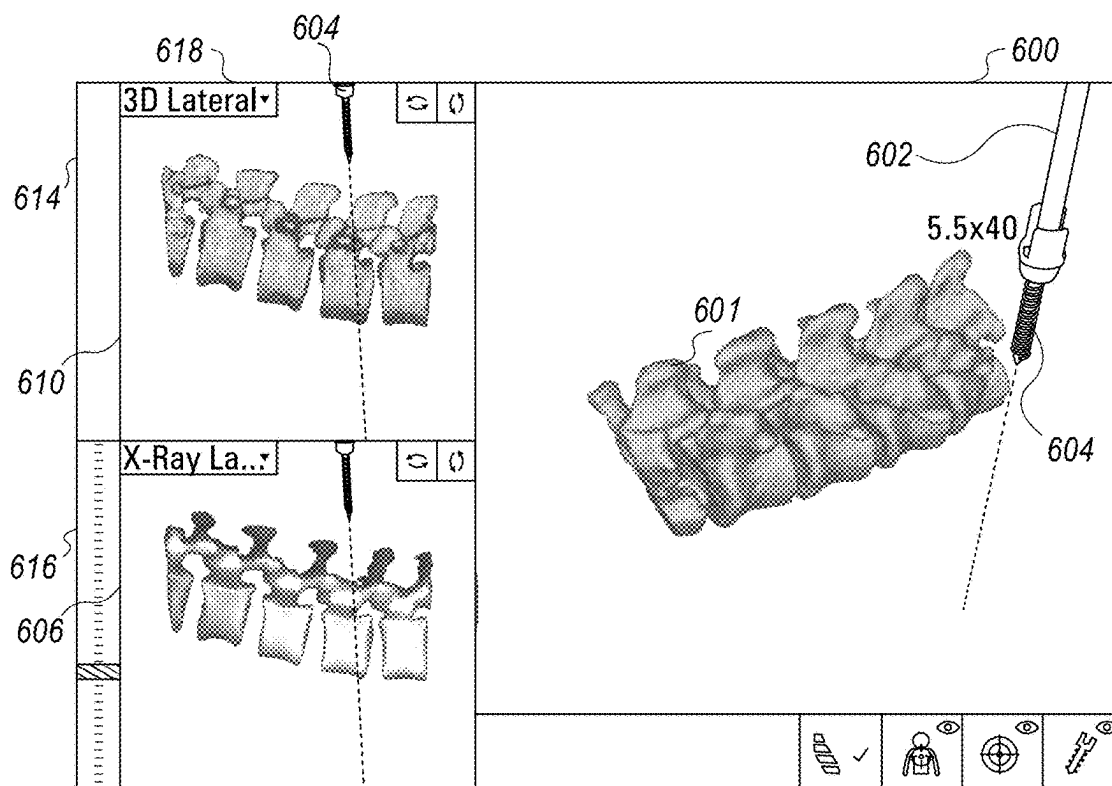


FIG. 6C

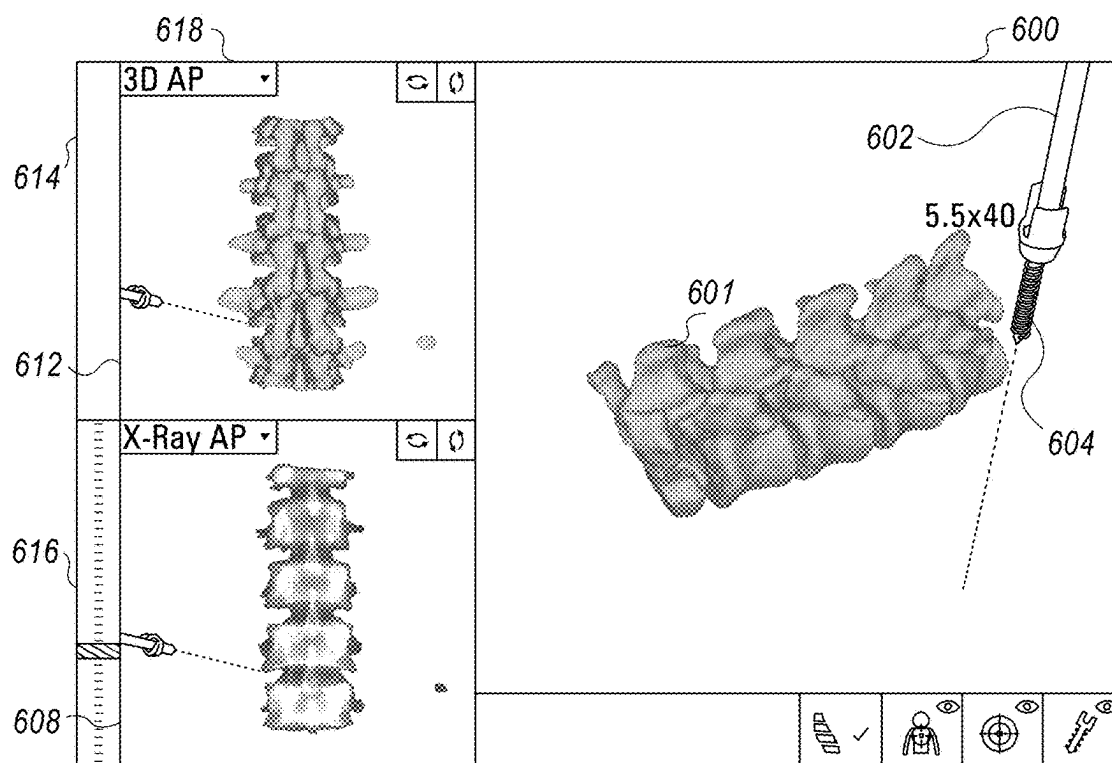


FIG. 6D

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METHODS FOR MEDICAL IMAGE VISUALIZATION

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/365,566, filed Aug. 4, 2023, which is a continuation of International PCT Application PCT/IB2023/054056, filed Apr. 20, 2023, which claims the benefit of U.S. Provisional Patent Application No. 63/333,128, filed Apr. 21, 2022 and of U.S. Provisional Patent Application No. 63/428,781, filed Nov. 30, 2022. The entire contents of each of the foregoing applications is incorporated herein by reference.

FIELD

The present disclosure generally relates to medical visualization, including systems and methods for generating (e.g., computing) three-dimensional (3D) models based on 2D or 3D medical images for use, among other considered medical usages, such as in image-guided surgery, and for generating 3D renderings based on the 2D or 3D medical images (e.g., for output on an augmented reality display).

BACKGROUND

Near-eye display devices and systems, such as head-mounted displays are commonly used in augmented reality systems, for example, for performing image-guided surgery. In this way, a computer-generated three-dimensional (3D) model of a volume of interest of a patient may be presented to a healthcare professional who is performing the procedure, such that the 3D model is visible to the professional engaged in optically viewing an anatomical portion of a patient who is undergoing the procedure.

Systems of this sort for image-guided surgery are described, for example, in Applicant's U.S. Pat. Nos. 9,928,629, 10,835,296, 10,939,977, PCT International Publication WO 2022/053923, and U.S. Patent Application Publication 2020/0163723. The disclosures of all of these patents and publications are incorporated herein by reference.

SUMMARY

In accordance with several embodiments, systems, devices and methods are described that provide enhanced or improved display of 3D models in connection with image-guided medical procedures. For example, the 3D models may be displayed with reduced noise and increased image quality. In some instances, the 3D models may not be displayed with background features (e.g., soft tissue is not displayed when it is bone tissue that is desired to be displayed). In some instances, the 3D models that are generated include implants or hardware (such as screws, rods, cages, pins, tools, instruments, etc.) but not certain types of tissue (e.g., soft tissue, nerve tissue) that is not the focus of the particular medical procedure. For example, the content of the 3D model or rendering that is generated for display may preferentially or selectively include only certain types of content that a clinical professional or other operator would want to see and not "background" content that the clinical professional or other operator does not need to see or does not want to see because it does not impact or affect the medical procedure (e.g., surgical or non-surgical therapeutic procedure or diagnostic procedure). In the example of

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a spinal surgical procedure, the background content that may be desired to be filtered out or selectively or preferentially not displayed may be soft tissue surrounding or between the vertebrae of the spine and the content that may selectively or preferentially be displayed is the bone tissue (and optionally, any hardware, implant or instruments or tools within the bone, such as screws, rods, cages, etc.).

In accordance with several embodiments, the systems, devices and methods described herein involve automatically segmenting the image(s) (e.g., 3D computed tomography images, 3D magnetic resonance images, other 2D or 3D images) received from an imaging device scan prior to applying one or more 3D model generation algorithms or processes. In accordance with several embodiments, the segmentation advantageously reduces noise around the bone structure and provides a better "starting point" for the 3D model generation algorithms or processes. The segmentation may involve segmenting the 3D image into multiple separate regions, sections, portions, or segments. An entire anatomical portion of a subject (e.g., an entire spine or entire other bone, such as a hip, knee, shoulder, ankle, limb bone, cranium, facial bone, jaw bone, etc.) may be segmented or a subportion of the anatomical portion may be segmented.

Several embodiments are particularly advantageous because they include one, several or all of the following benefits: (i) using a segmented image for 3D image rendering and/or model building to achieve a more accurate and less noisy 3D model and/or visualization of a patient anatomy (e.g., a portion of the patient anatomy); and/or (ii) using artificial intelligence based segmentation in 3D model building to support low-quality intra-operative scanners or imaging devices; and/or (iii) applying different threshold values to different portions of the medical image to achieve better visualization of the patient anatomy; and/or (iv) enhancing a 3D model building while using a model building algorithm which may receive only a single threshold by applying multiple thresholds and/or (v) allowing a user to select different thresholds to be applied on different portions of a medical image (e.g., to improve or optimize visualization).

In accordance with several embodiments, a system for improving display of 3D models in connection with image-guided medical procedures (e.g., surgical or non-surgical therapeutic and/or diagnostic procedures) comprises or consists essentially of a wearable device (e.g., head-mounted unit such as eyewear) including at least one see-through display configured to allow viewing of a region of a body of a patient through at least a portion of the display and at least one processor (e.g., a single processor or multiple processors) configured to perform actions (e.g., upon execution of stored program instructions on one or more non-transitory computer-readable storage media). For example, the at least one processor is configured to receive a three-dimensional (3D) image of the region of the body of the patient, the 3D image having intensity values; segment the 3D image to define at least one region of interest (ROI) of the region of the body (e.g., a portion of a spine or other bone associated with the medical procedure); determine at least one ROI intensity threshold value of the at least one ROI; determine a background intensity threshold value; and generate a 3D rendering of the 3D image.

The generation of the 3D rendering may include, in the at least one defined ROI of the 3D image, rendering based on intensity values of the at least one ROI that satisfy a lowest threshold value of the at least one ROI intensity threshold value and the background intensity threshold value; and, in a background region of the 3D image, rendering based on

intensity values of the background region that satisfy the background intensity threshold value. The background region of the 3D image may include a portion of the 3D image which is not an ROI.

The at least one processor may also be configured to cause the 3D rendering to be output to the display of the wearable device.

In some embodiments, the intensity values are values of 3D image voxels of the 3D image.

In some embodiments, the wearable device is a pair of glasses or other eyewear. In some embodiments, the wearable device is an over-the-head mounted unit, such as a headset.

In some embodiments, the wearable device is configured to facilitate display of 3D stereoscopic images that are projected at a distance to align with a natural focal length of eyes (or a natural convergence or focus) of a wearer of the wearable device to reduce vergence-accommodation conflict.

The 3D image may be a computed tomography image, a magnetic resonance image, or other 3D image generated by another 3D imaging modality.

In some embodiments, the at least one processor is configured to display the 3D rendering in alignment by performing registration of the 3D rendering with the body region of the patient (e.g., using one or more markers, such as retroreflective markers that can be scanned or imaged by an imaging device of the wearable device).

In some embodiments, the 3D rendering is a 3D model. The 3D rendering may be output for display as a virtual augmented reality image. The virtual augmented reality image may be projected directly on a retina of the wearer of the wearable device. The virtual augmented reality image may be presented in such a way that the wearer can still see the physical region of interest through the display.

In some embodiments, the at least one processor is configured to generate the 3D rendering by changing the determined intensity values of the 3D image into a value which does not satisfy the lowest threshold value.

In some embodiments, the background region of the 3D image includes soft tissue.

In some embodiments, the at least one processor is further configured to repeatedly adjust the at least one ROI intensity threshold value and the background feature intensity threshold value according to input from a user; and repeatedly generate a 3D rendering of the 3D image based on the adjusted values of the at least two intensity thresholds. Only one of the threshold values may be adjusted in some embodiments.

In accordance with several embodiments, a system for improving display of 3D models in connection with image-guided surgery comprises or consists essentially of a head-mounted unit including at least one see-through display configured to allow viewing of a region of a spine of a patient through at least a portion of the display and at least one processor configured to (e.g., upon execution of program instructions stored on one or more non-transitory computer-readable storage media): receive a three-dimensional (3D) image of the region of the spine of the patient, the 3D image having intensity values, segment the 3D image to define multiple regions of interest (ROI) of the spine (e.g., multiple vertebrae or multiple vertebral segments); determine a ROI intensity threshold value for each of the multiple ROIs of the spine; determine a background intensity threshold value; generate a 3D rendering (e.g., 3D model) of the 3D image; and cause the 3D rendering to be output to the display as a virtual augmented reality image. The generation

of the 3D rendering includes, in the multiple ROIs of the 3D image, rendering based on the determined ROI intensity values of the multiple ROIs that satisfy a lowest threshold value of the intensity threshold value and the background intensity threshold value and, in a background region of the 3D image, rendering based on intensity values of the background region that satisfy the background intensity threshold value. The background region of the 3D image includes a portion of the 3D image which is not an ROI.

In some embodiments, the intensity values are voxel values of the 3D image.

In some embodiments, the head-mounted unit is a pair of glasses or other form of eyewear, including eyewear without lenses. In some embodiments, the head-mounted unit is an over-the-head mounted unit (e.g. a headset).

In some embodiments, the display is configured to be displayed directly on a retina of a wearer of the head-mounted unit.

In some embodiments, the at least one processor is configured to display the 3D rendering in alignment by performing registration of the 3D rendering with the region of the spine.

In some embodiments, the at least one processor is configured to generate the 3D rendering by changing the determined intensity values of the 3D image into a value which does not satisfy the lowest threshold value.

In some embodiments, the at least one processor is further configured to repeatedly adjust the at least one ROI intensity threshold value and the background feature intensity threshold value according to input from a user and repeatedly generate a 3D rendering of the 3D image based on the adjusted values of the at least two intensity thresholds.

An embodiment of the present disclosure that is described hereinafter provides a computer-implemented method that includes obtaining a three-dimensional (3D) image of a region of a body of a patient, the 3D image having feature values. The 3D image is segmented to define one or more regions of interest (ROIs). The one or more regions of interest may include one or more portions of a bone or joint (e.g., a portion of a spine, individual vertebrae of a portion of a spine, a particular spinal segment, a portion of a pelvis or sacroiliac region, or other bone or joint). At least one region of interest (ROI) feature threshold is determined. A background feature threshold is determined. A 3D model is generated from the 3D image based on the determined at least one ROI feature threshold, the determined background feature threshold, and the segmentation. The 3D model is outputted for display to a user.

In some embodiments, the feature values are intensity values, the ROI feature threshold is an ROI intensity threshold, and the background feature threshold is a background intensity threshold.

In some embodiments, the intensity values are the 3D image voxel values, and the intensity thresholds are the 3D image voxel thresholds.

In an embodiment, the method further includes generating a 3D rendering of the 3D image, wherein the generation of the 3D rendering includes (i) in the at least one ROI of the 3D image, rendering based on feature values of the at least one ROI that satisfy the lowest threshold of the at least one ROI feature threshold and the background feature threshold, and (ii) in a background region of the 3D image, rendering based on feature values of the background region that satisfy the background feature threshold, wherein the background region of the 3D image includes a portion of the 3D image which is not an ROI. The 3D model is outputted for display to a user.

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In another embodiment, generating the 3D model includes using a 3D model generation algorithm and providing a feature threshold as input to the 3D model generation algorithm.

In some embodiments, the 3D model generation algorithm is a marching cubes algorithm.

In some embodiments, the provided feature threshold corresponds to one of the at least one ROI feature threshold.

In some embodiments, the provided feature threshold is determined based on the at least one ROI feature threshold and the background feature threshold.

In other embodiments, the provided feature threshold is selected as the lowest of the at least one ROI feature threshold value and the background feature threshold value.

In some embodiments, the generation of the 3D model includes selecting feature values of the 3D image satisfying the provided feature threshold and omitting portions of the background region of the 3D image with selected feature values from being an input to the generation of the 3D model.

In other embodiments, the omitting of portions of the background region includes changing the selected feature values of the portions of the background region into a value that does not satisfy the provided feature threshold.

In an embodiment, the determination of the at least one ROI feature threshold and of the background feature threshold includes receiving input values for at least one of: the at least one ROI feature threshold or the background feature threshold.

In some embodiments, the input values are received for the at least one ROI feature threshold and the background feature threshold.

In some embodiments, the input values are received from a user.

In some embodiments, the receiving of the input values from the user includes generating a Graphical User Interface (GUI) element to be displayed to the user, the GUI element allowing the user to adjust the input values, and the rendering and displaying of the 3D rendering is iteratively performed in correspondence to the user adjustment of the input values.

In an embodiment, in response to a request of the user, the 3D model is generated based on current input values for at least one of: the at least one ROI feature threshold or the background feature threshold.

In another embodiment, the input values are received for the at least one ROI feature threshold and the background feature threshold.

In some embodiments, the method further includes displaying a default 3D model based on default values for the at least one ROI feature threshold and the background feature threshold, and displaying the default 3D model to the user.

In some embodiments, the determining of the at least one ROI feature threshold includes, when multiple ROIs are considered, determining respective multiple ROI feature thresholds.

In some embodiments, the determining of the at least one ROI feature threshold includes setting a feature threshold value that differentiates bone from soft tissue.

In other embodiments, the determining of the background feature threshold includes setting a feature threshold value that differentiates metal from bone.

In an embodiment, the method further includes using the 3D model with an image-guided system. The image-guided system and methods described herein can be surgical, non-surgical or diagnostic.

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In another embodiment, the image-guided system is an augmented or mixed reality system including a direct see-through display, such as a Head Mounted Display (HMD). In some embodiments, the image-guided system is an augmented or mixed reality system that does not include a head mounted display/component or includes both head mounted and non-head mounted displays/components.

In some embodiments, the segmentation is performed based on one or more deep learning networks. In some embodiments, the deep learning networks are convolutional neural networks.

There is additionally provided, in accordance with another embodiment of the present disclosure, a computer-implemented method including obtaining a three-dimensional (3D) image of a region of a body of a patient, the 3D image having features. The 3D image is segmented to define one or more regions of interest (ROIs). At least one ROI feature threshold is determined. A background feature threshold is determined. A 3D rendering of the 3D image is generated, wherein the generation of the 3D rendering includes (i) in the at least one defined ROI of the 3D image, rendering based on feature values of the at least one ROI that satisfy the lowest threshold of the at least one ROI feature threshold and the background feature threshold, and (ii) in a background region of the 3D image, rendering based on feature values of the background region that satisfy the background feature threshold, wherein the background region of the 3D image includes a portion of the 3D image which is not an ROI. The 3D model is outputted for display to a user.

In some embodiments, the method further includes generating a 3D model based on the determined at least one ROI feature threshold, the determined background feature threshold, and the segmentation. The 3D model is displayed on a display.

In some embodiments, generating the 3D model includes using a 3D model generation algorithm and providing a feature threshold as input to the 3D model generation algorithm.

In an embodiment, the provided feature threshold is selected to be the lowest of the at least one ROI feature threshold and the background feature threshold.

In another embodiment, the provided feature threshold is the at least one ROI feature threshold, and the portions of the background region having selected feature values which do not satisfy the background intensity threshold are omitted.

In some embodiments, the determination of the at least ROI feature threshold and of the background feature threshold includes receiving input values for at least one of: the at least one ROI feature threshold or the background feature threshold.

In some embodiments, the method further includes displaying a default 3D rendering based on default values for the at least one ROI feature threshold and the background feature threshold, and displaying the default 3D rendering to the user.

There is further provided, in accordance with another embodiment of the present disclosure, a system for image-guided surgery (or other procedures such as non-surgical procedures and diagnostics), the system including at least one display, and at least one processor. The at least one processor is configured to (i) receive a three-dimensional (3D) image of a region of a body of a patient, the 3D image having intensity values, (ii) compute a 3D model of the region based on at least two intensity threshold values applied to the 3D image, and (iii) cause the computed 3D model to be output for display on the at least one display.

In some embodiments, the at least one display includes a see-through display configured to allow viewing of a body of a patient through at least a portion of the display.

In some embodiments, the see-through display is included in a head-mounted unit. The head-mounted unit may provide elimination of attention shift and a reduced (e.g., minimized) line of sight interruption.

In other embodiments, the computed 3D model is displayed in alignment with the body of the patient as viewed through the see-through display. The 3D model may be provided in a 3D stereoscopic view. The virtual images may be projected at 50 cm in order to align with a normal focal distance of the eyes so as to reduce “vergence-accommodation conflict”, which can result in fatigue, headache, or general discomfort. The line of sight may be at approximately 30 degrees.

In other embodiments, the at least one processor is configured to display in alignment by performing registration of the 3D model with the body of the patient.

In some embodiments, the intensity values are the 3D image voxel values.

In an embodiment, the at least one display includes a stationary display. In some embodiments, the display is a display of a head-mounted device, including but not limited to glasses, goggles, spectacles, visors, monocle, other eye-wear, or over-the-head headset. In some embodiments, the head-mounted displays are not used or used together with stand-alone displays, such as monitors, portable devices, tablets, etc. The display may be a hands-free display such that the operator does not need to hold the display.

The head-mounted device may alternatively be a wearable device on a body part other than the head (e.g., a non-head-mounted device). The head-mounted device may be substituted with an alternative hands-free device that is not worn by the operator, such as a portal, monitor or tablet. The display may be a head-up display or heads-up display.

In an embodiment, the at least one processor is further configured to segment the 3D image to define one or more regions of interest (ROIs), and the computation of the 3D model is further based on the segmentation.

In another embodiment, the at least two intensity threshold values include at least one ROI threshold intensity value and a background threshold intensity value, and the at least one processor is further configured to (i) compute the 3D model by selecting intensity values of the 3D image satisfying the lowest intensity threshold value of the at least one ROI threshold intensity value, and (ii) omit portions of a background region of the 3D image with intensity values between the lowest intensity threshold value of the at least one ROI threshold intensity values and the background threshold intensity value from being an input to the generation of the 3D model.

In some embodiments, the at least one processor is configured to generate the 3D model by changing the selected intensity values of the 3D image into a value which does not satisfy the lowest intensity threshold value.

In some embodiments, the at least two intensity thresholds include at least one region of interest (ROI) intensity threshold to be applied to one or more ROIs and a background intensity threshold to be applied to at least a background region of the 3D image, and the background region of the 3D image includes a portion of the 3D image which is not an ROI.

In other embodiments, the at least one processor is further configured to (i) repeatedly adjust the at least two intensity thresholds according to input from a user, and (ii) repeatedly

generate a 3D rendering of the 3D image based on the adjusted values of the at least two intensity thresholds.

In an embodiment, a method is provided of generating at least one of a 3D model and images for display.

In an embodiment, a system is provided, that includes at least one processor configured for facilitating generation of at least one of a 3D model and images, the system further including a display device configured to display the at least one of a 3D model and images.

Also described and contemplated herein is the use of any of the apparatus, systems, or methods for the treatment of a spine through a surgical intervention.

Also described and contemplated herein is the use of any of the apparatus, systems, or methods for the treatment of an orthopedic joint through a surgical intervention, including, optionally, a shoulder, a knee, an ankle, a hip, or other joint.

Also described and contemplated herein is the use of any of the apparatus, systems, or methods for the treatment of a cranium through a surgical intervention.

Also described and contemplated herein is the use of any of the apparatus, systems, or methods for the treatment of a jaw through a surgical intervention.

Also described and contemplated herein is the use of any of the apparatus, systems, or methods for diagnosis of a spinal abnormality or degeneration or deformity.

Also described and contemplated herein is the use of any of the apparatus, systems, or methods for diagnosis of a spinal injury.

Also described and contemplated herein is the use of any of the apparatus, systems, or methods for diagnosis of joint damage.

Also described and contemplated herein is the use of any of the apparatus, systems, or methods for diagnosis of an orthopedic injury.

In accordance with several embodiments, any of the methods described herein may include diagnosing and/or treating a medical condition, the medical condition comprising one or more of the following: back pain, spinal deformity, spinal stenosis, disc herniation, joint inflammation, joint damage, ligament or tendon ruptures or tears.

In accordance with several embodiments, a method of presenting one or more images on a wearable display is described and/or illustrated herein during medical procedures, such as orthopedic procedures, spinal surgical procedures, joint repair procedures, joint replacement procedures, facial bone repair or reconstruction procedures, ENT procedures, cranial procedures or neurosurgical procedures.

For purposes of summarizing the disclosure, certain aspects, advantages, and novel features of embodiments of the disclosure have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment of the disclosure disclosed herein. Thus, the embodiments disclosed herein may be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught or suggested herein without necessarily achieving other advantages as may be taught or suggested herein. The systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible for the desirable attributes disclosed herein. The methods summarized above and set forth in further detail below describe certain actions taken by a practitioner; however, it should be understood that they can also include the instruction of those actions by another party. Thus, actions such as “adjusting a threshold” include “instructing the adjustment of a threshold.”

The present disclosure will be more fully understood from the following detailed description of the embodiments thereof, taken together with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting features of some embodiments are set forth with particularity in the claims that follow. The following drawings are for illustrative purposes only and show non-limiting embodiments. Features from different figures may be combined in several embodiments. It should be understood that the figures are not necessarily drawn to scale. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated.

FIG. 1 is a schematic, pictorial illustration of an example system for image-guided surgery or other medical intervention utilizing a head-mounted display, in accordance with an embodiment of the disclosure.

FIG. 2 is a flowchart of steps performed to generate a 3D rendering and/or a 3D model based on a 3D medical image of at least a portion of a body of a patient, in accordance with an embodiment of the disclosure.

FIGS. 3A-3C show graphical user interface displays that include 3D renderings based on processed 3D images that facilitate the determination of feature thresholds used for computing respective 3D models, in accordance with embodiments of the disclosure.

FIG. 4 is a graphical user interface display that includes a generated 3D model of a volume of interest of a patient, the 3D model computed based on the 3D image used to generate the 3D rendering of FIG. 3C, in accordance with an embodiment of the disclosure.

FIG. 5 is a flowchart of steps performed to generate a three-dimensional (3D) model based on a 3D image of a portion of a body of a patient using a selected threshold value, in accordance with an embodiment of the disclosure.

FIGS. 6A-6D are example graphical user interface displays that include various views of a patient's spine, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

Overview

A three-dimensional (3D) model of a volume of interest of a patient may include irrelevant or undesired information around the volume of interest or around the model, such as irrelevant or undesired objects or tissue and/or image noise. A user may adjust or cause adjustment of a threshold (e.g., radiodensity threshold value such as using a Hounsfield Unit scale) of an image (e.g., a CT image or other 2D or 3D medical image), to suppress the irrelevant or undesired information. However, reducing irrelevant or undesired information this way may cause loss of some of the relevant or desired information. For example, a 3D image may include metal objects characterized by very high radiodensity values and bone characterized by lower radiodensity values. Setting a high enough threshold may show the metal objects and remove noise but may well erode or remove some bone structure features in the image.

In accordance with several embodiments, a pre-processing stage may be included to automatically segment the medical image (e.g., 3D CT scan) prior to generation of a 3D model (e.g., generating a 3D model by applying one or more 3D model generation algorithms or processes). The segmen-

tation may advantageously reduce the noise around the bone structure and provide a better "starting point" for the 3D model generation. The segmentation may be applied to the entire medical image or to one or more portions, regions or segments of the medical image.

Image-guided surgery employs tracked surgical instruments and images of the patient anatomy in order to guide the procedure. In such procedures, a proper visualization of regions of interest of the patient anatomy, including tissue of interest, is of high importance. Enhanced or improved visualization may also advantageously provide increased user adoption and enhance user experience.

In particular, during image-guided surgery or other medical intervention (e.g., procedures employing augmented reality technology, virtual reality technology or mixed-reality technology), the 3D model of a volume of interest of the patient upon whom the surgery or other medical intervention (including diagnostic and/or therapeutic intervention) is being performed may be aligned with tools or devices used during the surgery or intervention and/or with the anatomical portion of the patient undergoing surgery or other medical intervention. Such 3D model may be derived from a 3D medical image, such as a 3D CT image or 3D MR image. In accordance with several embodiments, a proper modeling based on the 3D medical image is important for providing the user with the required and accurate information for navigating a medical tool on anatomical images derived from the 3D medical image (e.g., slices, 3D model and/or 2D images) and for accurate and/or proper augmentation (e.g., in alignment) of the 3D model with the optically viewed (e.g., with a near-eye display device) anatomical portion of the patient when augmented-reality is used.

Some embodiments of the present disclosure provide computer-implemented methods for generating a 3D model from at least one 3D medical image. The methods include obtaining one or more 3D images of at least a region of a body of a patient (e.g., including a cervical spine region, a thoracic spine region, a lumbosacral spine region, a cranial region, an ENT-associated region, a facial region, a hip region, a shoulder region, a knee region, an ankle region, a joint region, or a limb region), the 3D image having one or more feature values, such as image intensity values on a gray-scale, RGB values, gradients, edges and/or texture. In some embodiments, the 3D image is first segmented to include or identify one or more regions of interest (ROIs). At least one ROI feature threshold may be determined. At least one background feature threshold may also be determined. In some embodiments, a 3D model is generated from the 3D image, based on the determined at least one ROI feature threshold, the determined at least one background feature threshold, and the segmentation. The 3D model may then be saved for use in image-guided surgery and/or systems, e.g., to be displayed to a user on a display of a wearable (e.g., head-mounted or near-eye or direct see-through) display device and/or on a separate or stationary display monitor or device (e.g., a standalone or separate portal, tablet or monitor).

In some embodiments, the 3D image is processed to only show the at least one ROI and to only show background information above the background feature threshold. This may be performed, for example, by segmenting the image and applying different feature thresholds to different portions of the image. In such embodiments, irrelevant or undesired background information can be suppressed using a high background feature threshold, without desired features inside the at least one ROI being eroded.

As noted above, the techniques disclosed herein may advantageously present (e.g., cause to be displayed) maximum or an increased amount of information obtained from ROIs of the 3D medical image and filter out some of the background of the 3D medical image, which may include, for example, noise or information of less interest. Accordingly, in some embodiments, the at least one processor may be configured to apply the lowest feature threshold to the ROIs (e.g., even if a specific ROI's feature threshold is not or different than the lowest feature threshold). In some embodiments, the at least one processor may be configured to apply the background feature threshold to the entire image while determining in each region or segment of the image the lowest feature threshold applied to the region or segment as the specific region or segment feature threshold. For example, for a specific ROI, a background feature threshold and an ROI feature threshold are applied. One may expect that the ROI feature threshold would be the lowest but in case the background feature threshold is the lowest, then the background feature threshold value would determine the threshold value for the specific ROI. Specifically, a background feature (e.g., intensity) threshold may be set to a higher value than any of one or more ROI feature (e.g., intensity) thresholds, for example. As a result, the processed 3D image (e.g., processed via volume rendering) may show only information contained in the at least one ROI and background information (e.g., objects) of sufficient intensity, while removing irrelevant or undesired background information. The processed 3D image may show in an ROI bone structure with metal surgical objects in the background, without showing irrelevant anatomy (e.g., other bones or other type of tissue) or an image suffering from background image noise.

In some embodiments, at least two feature thresholds are provided that include at least one region of interest (ROI) feature threshold to be applied to the one or more ROIs, and a background feature threshold to be applied to a background region of the 3D image. The background region of the 3D image may include a portion of the 3D image which is not an ROI or the portion or all portions of the 3D image which are not an ROI. By suppressing the values of the background regions (e.g., values of 3D image voxels) which do not satisfy the background feature threshold, an effective unique threshold may be obtained. This suppression step may be used for generating a 3D model using an algorithm that accepts only one threshold value (e.g., marching cubes algorithm), while at the same time maximizing or otherwise increasing information obtained from ROIs of the 3D medical image and filtering out some of the background of the 3D medical image, which may be an area of less interest.

In some embodiments of the computer-implemented method, before the 3D model is computed, a 3D rendering based on a processed 3D image (e.g., a segmented image) is displayed to a user, such as a surgeon or other healthcare professional. The user may view the 3D rendering on a monitor or a display (e.g., a graphical user interface display on a monitor or on a near-eye display of a head-mounted, direct see-through, or near-eye display device). The graphical user interface display may allow the user to adjust the at least one ROI intensity threshold and/or the background intensity threshold, to thereby modify the look of the 3D rendering. This way, the user can advantageously optimize the rendered 3D image by selecting and/or identifying optimal and/or desired feature thresholds, based on which the 3D model will be computed.

The 3D rendering may be made of the original image voxels, or may be a result of further image processing steps

applied to the 3D image, such as surface ray casting. Further image processing of the 3D image beyond producing a 3D rendering using the native 3D image voxels can have advantages. For example, using ray casting may facilitate interpolation between, e.g., voxel intensity values, to produce a smoother look. Also, some methods such as surface ray cast rendering use only a portion of the information included in the native 3D image (e.g., use only voxels encountered by rays, provided the voxels have feature values such as intensities above a threshold). In accordance with several embodiments, using only a subset of the voxels minimizes or otherwise advantageously reduces computation effort and time required to generate a 3D rendering. This reduction may be particularly important as the user manipulates the view of the rendering (e.g., views the rendering from different points of view), which requires repeated computations of the 3D rendering.

In accordance with several embodiments, the 3D rendering generation process comprises the steps of:

In the at least one ROI of the 3D image, rendering based on feature values of the at least one ROI which satisfy the lowest threshold of the at least one ROI feature threshold and the background feature threshold.

In a background region of the 3D image, rendering based on feature values of the background region which satisfy the background feature threshold, wherein the background region of the 3D image comprises a portion of the 3D image which is not an ROI.

As noted above, to generate the 3D model, two thresholds (e.g., the ROI feature threshold and the background feature threshold) may be determined by a user (or automatically by at least one processor, e.g., based on accumulated data or based on trained machine learning algorithms or neural networks). However, some 3D model generation algorithms (e.g., surface building algorithms, such as the Marching Cubes algorithm), may receive as input only a single threshold value. In accordance with several embodiments, the techniques disclosed herein advantageously solve the limitation of only a single allowed input threshold by:

Using the lowest ROI feature threshold value with the algorithm (e.g., running the Marching Cubes algorithm) to delineate ROI regions.

Visualizing only ROI regions with feature values higher than a lowest ROI feature threshold.

Visualizing only portions of any other regions, including background regions, with feature values higher than a background feature threshold.

In accordance with several embodiments, background information (e.g., voxel values) having feature values higher than the ROI feature threshold but lower than the background feature threshold is removed (e.g., in order not to model and display irrelevant or undesired information). In some embodiments, such removal is performed by determining the value of such voxels to a value which does not satisfy the ROI feature threshold(s), e.g., equal to or lower than the ROI feature threshold (e.g., ROI threshold-C) while C is a constant (non-negative number): $C > 0$. In some embodiments, the subtracting of such a constant may allow further smoothing of the visualization (e.g., the 3D model).

As described above, using two thresholds, called herein TH ROI and TH Background, may allow showing ROI information and some selected background objects, respectively. If ROI information has to be visualized has to meet multiple thresholding conditions (e.g., multiple tissue types have to be visualized, such as fat, muscle and bone), then two or more (e.g., multiple) ROI thresholds (e.g., one for each ROI type), $TH(j)$ ROI, $j=1, 2, \dots$, may be defined and

used in segmentation, volume rendering and/or generating a 3D model. In some embodiments when only one threshold may be input to a 3D model generation algorithm, then the lowest of {TH(j)ROI} may be used.

In one embodiment, a trained neural-network (NN) based segmentation algorithm is provided. The disclosed segmentation algorithm classifies different regions of 3D image as ROI and background and delineates one from another. For example, in orthopedic image-guided applications (such as surgery or other procedures, such as non-surgical or diagnostic procedures), the disclosed segmentation algorithm may be trained to classify vertebrae column and the Iliac and Sacrum as "Spine". According to some embodiments, the vertebrae, each vertebra, the Iliac and/or the Sacrum may be segmented separately. According to some embodiments, screws, implants and/or inter-bodies included in the scan may also be classified by the algorithm as "Spine" (or the desired category to be shown). Soft tissues, the rib cage and instruments such as clamps, retractors and rods that are included in the scan may be classified as "background." In some implementations, the segmentation of the spine is done on the whole spine anatomy included in the scan. In some implementations, the segmentation of the spine is performed on portions of the spine anatomy (e.g., just the lumbosacral region, just the lumbar region, just the thoracic region, just the sacral region, just the sacroiliac region, just the cervical region, or combinations thereof) or on individual vertebrae, and/or on anatomical portions of individual vertebrae. The segmentation may further be used for creating a virtual 3D model of the scanned spine area, leaving out, for example, soft tissue, ribs and rods. However, in some embodiments, such left-out elements may be segmented and classified as ROI, such as ribs.

In some embodiments, a segmentation method is applied for generating a 3D model without resorting to a subsequent model building step and algorithm. To this end, the segmentation may use ROI and background feature threshold values in the delineation process. In some embodiments, the segmentation algorithm uses the minimal ROI feature threshold value to segment a first group of regions (e.g., bone regions) in the 3D image, with values above the minimal ROI feature threshold value. Then, the segmentation algorithm uses the background intensity threshold to further segment a second group of regions in the 3D image, with values above the background feature threshold value. A processed image is then generated that visualizes only first and second regions obtained by the segmentation algorithm. If the spatial resolution of the segmentation algorithm is sufficiently high, as well as that of the native 3D image, the segmentation algorithm may be sufficient to create a 3D model (e.g., without resorting to surface building algorithms, such as the Marching Cubes algorithm).

In some embodiments, the at least one processor is configured to use the segmentation to generate a digitally reconstructed radiograph (DRR) of one or more ROIs only. To this end, the at least one processor may be configured to omit the background region, for example, by setting voxel feature values in the medical 3D image (e.g., an input 3D Digital Imaging and Communications in Medicine (DICOM) image) segmented as background to zero. The DRRs may be generated by summarizing feature values in one dimension or by using methods such as Siddon's algorithm published in 1985 by Robert L. Siddon. For example, in Siddon's algorithm, the index of each voxel along a certain projection ray and the interesting length of that ray within that voxel may be computed by four multiplications. Refinements or modifications to Siddon's algorithm, or alternative

algorithms (e.g., for calculating radiological paths through pixel or voxel spaces), may also be used.

A further embodiment provides an augmented-reality system for image-guided surgery (or other procedures such as non-surgical procedures and diagnostics), the system comprising (i) a see-through augmented-reality display configured to allow viewing of a body of a patient through at least a portion of the display by the user (e.g., wearing a head-mounted unit), and at least one processor, which is configured to receive the 3D image of a region of the body of the patient and compute a 3D model of the region based on at least two intensity thresholds applied to the 3D image, and present the computed 3D model on the display in alignment with the body of the patient viewed through the display, e.g., for better spatial awareness.

In some embodiments, accurate and/or proper augmentation (e.g., alignment) of a navigated medical tool with images derived from the medical image (e.g., slices, 3D model, 2D images and/or DRR images) and, optionally, accurate and/or proper augmentation of the 3D model with an optically viewed anatomical portion is facilitated by a registration step of the 3D medical image, e.g., via artificial fiducials in the 3D medical image and by utilizing a tracking system. Specifically, when using a direct see-through or head-mounted display, accurate and/or proper augmentation of the 3D model with an optically viewed anatomical portion as viewed from the point of view of a professional wearing the head-mounted display, may be achieved by utilizing a tracking device for tracking at least the position of the head-mounted display relative to the anatomical portion. In some embodiments the tracking system may be mounted on the head-mounted display device. Systems of this sort using augmented-reality for image-guided surgery (or other procedures such as non-surgical procedures and diagnostics), are described, for example, in the above incorporated Applicant's U.S. Pat. Nos. 9,928,629, 10,939,977, PCT International Publication WO 2022/053923, and U.S. Patent Application Publication 2020/0163723.

The disclosure describes, for example, a method of generating a 3D model and/or images for display (e.g., to facilitate image-guided surgery using a direct see-through or head-mounted augmented reality display device) such as described and/or illustrated hereinafter. The disclosure further describes a system comprising at least one processor for generating a 3D model and/or images for display, and a display device configured to display the 3D model and/or images (e.g., to facilitate image-guided surgery using a head-mounted augmented reality display device) such as described and/or illustrated hereinafter.

Finally, as noted above, one common type of 3D medical image in use is a 3D CT image. Another type of 3D medical image that can be used is a 3D MM image. The method can use, mutatis mutandis, 2D or 3D medical images derived from other imaging modalities, such as ultrasound, positron emission tomography (PET), and optical coherence tomography (OCT). Therefore, the above-described methods may be applied to other types of medical acquisitions (e.g., medical scans or images), including electroencephalogram (EEG) imaging, and to other areas or organs of the body, such as skull or cranium, knees, hips, shoulders, sacroiliac joints, ankles, ear, nose, throat, facial regions, elbows, other joint regions or orthopedic regions, gastrointestinal regions, etc. Accordingly, the feature values may be selected as any relevant physiological or electrophysiological values, such as blood and neural activation velocities, respectively, or speeds as derived from the image features.

The systems and methods described herein may be used in connection with surgical procedures, such as spinal surgery, joint surgery (e.g., shoulder, knee, hip, ankle, other joints), orthopedic surgery, heart surgery, bariatric surgery, facial bone surgery, dental surgery, cranial surgery, or neurosurgery. The surgical procedures may be performed during open surgery or minimally-invasive surgery (e.g., surgery during which small incisions are made that are self-sealing or sealed with surgical adhesive or minor suturing or stitching. However, the systems and methods described may be used in connection with other medical procedures (including therapeutic and diagnostic procedures) and with other instruments and devices or other non-medical display environments. The methods described herein further include the performance of the medical procedures (including but not limited to performing a surgical intervention such as treating a spine, shoulder, hip, knee, ankle, other joint, jaw, cranium, etc.).

System Description

FIG. 1 is a schematic, pictorial illustration of an example system 10 for image-guided surgery or other medical intervention utilizing a head-mounted display such as a direct see-through display, in accordance with an embodiment of the disclosure. FIG. 1 is a pictorial illustration of the system as a whole, including showing, by way of example, a head-mounted display unit 28 in a form of eye-glasses, spectacles, or other eyewear that is used in system 10. Alternative types of a head-mounted unit, such as ones based on a helmet, visor, or over-the-head mounted unit that the surgeon wears, may be used in system 10. Alternatively, other wearable or hands-free devices, units or displays may be used.

In accordance with several embodiments, system 10 is applied in a medical procedure on a patient 20 using image-guided intervention. In this procedure, a tool 22 is inserted via an incision (e.g., minimally-invasive or self-sealing incision) in the patient's back in order to perform a surgical intervention. Alternatively, system 10 and the techniques described herein may be used, mutatis mutandis, in other surgical or non-surgical medical or diagnostic procedures.

In the pictured embodiment, a user of system 10, such as a healthcare professional 26 (e.g., a surgeon performing the procedure), wears the head-mounted display unit 28. In various embodiments, the head-mounted display unit 28 includes one or more see-through displays 30, for example as described in the above-mentioned U.S. Pat. No. 9,928,629 or in the other patents and applications cited above that are incorporated by reference. Such see-through displays may include an optical combiner that is controlled by a computer processor (e.g., a computer processor 32 in a central processing system 50 and/or a dedicated computer processor in the head-mounted display unit 28) to display an augmented-reality image to the healthcare professional 26. The image is presented on a see-through display 30 (e.g., as a stereoscopic near-eye display) such that a computer-generated image is projected in alignment with the anatomy of the body of patient 20 that is visible to professional 26 through a portion of the see-through display. In some embodiments, one or more projected computer-generated images include a virtual image of tool 22 overlaid on a virtual image of at least a portion of the patient's anatomy (such as slices, the 3D model or 2D images derived from the 3D medical image, e.g., of a portion of the spine). Specifically, a portion of the tool 22 that would not otherwise be visible to the healthcare

professional 26 (for example, by virtue of being hidden by the patient's anatomy) is included in the computer-generated image.

In image-guided surgery or in surgeries utilizing augmented reality systems, a bone-anchoring device may be used as a fiducial marker or may be coupled with such a marker, for indicating the patient's body location in a coordinate system. In system 10, for example, an anchoring device is coupled with a marker that is used to register a region of interest (ROI) of the patient body with a CT scan or other medical images of the ROI (e.g., preoperative, or intraoperative CT or fluoroscopic image). During the procedure, a tracking system (e.g., an IR tracking system of the head-mounted display unit 28) tracks a marker mounted on the anchoring device and a tool that includes a tool marker. Following that, the display of the CT or other medical image data (which may include, for example, a model generated based on such data) on a near-eye display may be aligned with the professional's actual view of the ROI based on this registration. In addition, a virtual image of the tool 22 may be displayed on the CT or other 3D image model based on the tracking data and the registration. The user (e.g., professional 26) may then navigate the tool 22 based on the virtual display of the tool 22 with respect to the patient image data, and optionally, while the 3D model is aligned with the professional's view of the patient or ROI.

According to some aspects, the 3D image or model presented on see-through display 30 is aligned with the patient's body. According to some aspects, allowed alignment error (or allowed misalignment) may not be more than about two to three mm or may be less than four mm or less than five mm or about one to two mm or less than one mm. In one embodiment, the allowed alignment error is less than one mm. In order to account for such a limit on error in alignment of the patient's anatomy with the presented images, the position of the patient's body, or a portion thereof, with respect to the head-mounted unit may be tracked. According to some aspects, the 3D model presented on see-through display 30 is misaligned with the patient's body (e.g., having a misalignment greater than the allowed misalignment).

A patient marker 60 attached to an anchoring implement or device such as a clamp 58 or a pin, for example, may be used for this purpose, as described further hereinbelow. In some embodiments, a registration marker, such as registration marker 38, may be used for registering the 3D image with the patient anatomy, e.g., in a preceding registration procedure. In some embodiments registration marker 38 may be removed once the registration is complete. According to some embodiments, patient marker 60 and registration marker 38 may be combined into a single marker. Anchoring devices, markers and registration systems and methods of this sort for image-guided surgery are described, for example, in the above incorporated U.S. Pat. Nos. 9,928,629, 10,835,296 and 10,939,977, and in addition in Applicant's U.S. Patent Application Publication 2021/0161614, U.S. Patent Application Publication 2022/0142730, U.S. Patent Application Publication 2021/0386482, U.S. Patent Application Publication 2023/0009793, and PCT International Publication WO 2023/026229, all of which are incorporated herein by reference.

When an image of tool 22 is incorporated into the image that is displayed on head-mounted display unit 28, the position of the tool 22 with respect to the patient's anatomy should be accurately reflected. For this purpose, the position of the tool 22 or a portion thereof (e.g., a tool marker 40) is tracked by system 10 (e.g., via a tracking system built into

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the head-mounted display unit 28). In some embodiments, it is desirable to determine the location of the tool 22 with respect to the patient's body such that errors in the determined location of the tool 22 with respect to the patient's body are typically less than three mm or less than two and a half mm or less than two mm.

In some embodiments, head-mounted display unit 28 includes a tracking sensor 34 to facilitate determination of the location and orientation of head-mounted display unit 28 with respect to the patient's body (e.g., via patient marker 60) and/or with respect to tool 22 (e.g., via a tool marker 40). Tracking sensor 34 can also be used in finding the position and orientation of tool 22 (e.g., via tool marker 40) with respect to the patient's body (e.g., via patient marker 60). In one embodiment, tracking sensor 34 comprises one or more image-capturing or image-acquiring devices, such as a camera (e.g., infrared camera, visible light camera, and/or an RGB-IR camera), which captures or acquires images of patient marker 60 and/or tool marker 40, and/or other landmarks or markers. The tracking sensor 34 may comprise an optical tracking device, an RFID reader, an NFC reader, a Bluetooth sensor, an electromagnetic tracking device, an ultrasound tracking device, or other tracking device.

In the pictured embodiment, system 10 also includes a tomographic imaging device, such as an intraoperative computerized tomography (CT) scanner 41. Alternatively or additionally, processing system 50 may access or otherwise receive tomographic data from other sources; and the CT scanner 41 itself is not necessarily an essential part of the present system 10 (e.g., it is optional). Regardless of the source of the tomographic data, processor 32 computes a transformation over the ROI so as to register the tomographic images with images and information (e.g., a depth map) that are displayed on head-mounted display unit 28. The processor 32 can then apply this transformation in presenting at least a part of the tomographic image on see-through display 30 in registration with the ROI viewed through the see-through display 30. In the disclosed technique, the processor 32 can generate from the tomographic images, collectively named 3D image, a 3D model, and register the 3D model with the ROI, e.g., viewed through the see-through display 30.

In accordance with several implementations, in order to generate and present an augmented reality image on see-through display 30, processor 32 computes the location and orientation of head-mounted display unit 28 with respect to a portion of the body of patient 20 (e.g., the patient's back or a portion thereof). In some embodiments, processor 32 also computes the location and orientation of tool 22 with respect to the patient's body. In one embodiment, a processor which is integrated within the head-mounted display unit 28, may perform these functions or a portion of these functions. Alternatively or additionally, processor 32, which is disposed externally to the head-mounted display unit 28 and which may be in wireless communication with the head-mounted display unit 28, may be used to perform these functions or a portion of these functions. Processor 32 can be part of processing system 50, which additionally includes an output device 52 (e.g., a display, such as a monitor) for outputting information to an operator of the system, memory, and/or an input device 54 (such as a pointing device, a keyboard, a mouse, a trackpad, a pedal or touch-screen display, etc.) to allow the operator (e.g., professional 26 or an assistant to professional 26) to input data into the system 10.

In general, in the context of the present description, when a computer processor is described as performing certain

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steps, these steps may be performed by external computer processor 32 and/or a computer processor that is integrated within the head-mounted display unit 28. The processors described herein (e.g., processor 32) may include a single processor or multiple processors (e.g., parallel processors running in parallel to reduce computing time). If multiple processors are used, they may be communicatively coupled to each other over a network (e.g., wired or wireless communications network). The processor or processors carry out the described functionality under the control of suitable software, which may be downloaded to system 10 in electronic form, for example over a network, and/or stored on tangible, non-transitory computer-readable media, such as electronic, magnetic, or optical memory. In some embodiments, processing system 50 may be integrated in head-mounted display unit 28. In some embodiments, system 10 may include a see-through and/or augmented reality display which is not head-mounted. In such systems the display may be mounted to patient 20 or to the operating table and positioned such that professional 26 may be able to view patient 20 anatomy and/or operation site through at least a portion of the display. The display position may be adjustable.

In some embodiments, the 3D image, 3D rendering and 3D model of the present disclosure may be displayed while allowing user interaction as described herein on a display which is not a see-through display and/or a head-mounted display, such as a display of a workstation, a personal computer, a terminal, a tablet or a mobile device.

Visualizing a 3D Medical Image

FIG. 2 is a flowchart of steps performed to visualize a 3D medical image of a portion of a body of a patient, in accordance with an embodiment of the disclosure. The process can run automatically without user input, by one or more processors running all of the steps. Optionally, a user, such as the professional 26, may give input in thresholds obtaining step 206 and/or optical image cropping step 218, as described below. In some embodiments, the processor can be more than one processor, such that various steps are performed by different processors or processing units. In some embodiments, a single processor runs all of the steps. Thus, when a processor is mentioned herein, it may be referring to a single processor or multiple processors or processing units.

The process begins by the processor obtaining a 3D medical image (e.g., a CT scan) of a volume of the body portion, at a medical image obtaining step 202.

In general, e.g., depending on the clinical application, various medical image formats or modalities may be considered, such as MRI, PET or ultrasound images. In the contexts of this disclosure, the term "obtaining" may refer to, inter alia, acquiring, receiving, and/or accessing the 3D image and does not necessarily require the system to include the imaging device and/or the method to include the image acquiring operation or step.

Next, the processor segments the medical image to delineate at least one ROI (e.g., a spine section or region) from the rest of the imaged volume, at an image segmentation step 204. Various segmentation methods may be used, such as one that uses a trained convolutional neural network (CNN, such as U-net or V-net).

In some embodiments, a 3D image of the spine is segmented into spinal regions (e.g., cervical, thoracic, lumbar, sacral). In some embodiments, a 3D image of the spine is segmented into individual 3D vertebrae. In some embodi-

ments, a 3D image of the spine is further segmented to indicate anatomical regions or portions of a vertebra (for example, spinous process, articular processes, transverse processes, vertebral body, centrum, posterior vertebral arch or neural arch). Bones other than the bones of the spine or other tissue may also be segmented into various regions, into individual components, and/or into anatomical portions of individual components. Such segmentation, although requiring more computation resources, may provide a better segmentation of the spine or any other such complex structure. This segmentation operation can advantageously be carried out by deep learning techniques, using one or more trained CNNs. The sacrum and ilium may be segmented in this manner, as well, using a separate neural network from the neural network used for the vertebrae or the same neural network.

In some embodiments, a combination of three networks may be used for this purpose. The networks are fully convolutional networks, e.g., based on the U-Net architecture.

The processor may obtain at least one ROI threshold and background threshold levels or values, TH1 ROI and TH2 background, respectively, at thresholds obtaining step 206. Examples of different ROI intensity threshold and background intensity threshold levels used are described in connection with FIGS. 3A-3C below. In the examples given by FIGS. 3A-3C, the ROI and background thresholds are HU thresholds that govern CT 3D visualization of mainly bone and metal. In these figures, of an orthopedic application concerning a spine, the main aim of the background threshold is to visualize metal elements that are in the background section (e.g., external to the spine) such as clamps and retractors without adding irrelevant or undesired information (e.g., background noise, such as noisy voxel levels, having a radiodensity value which does not satisfy the background threshold).

In another example, relevant to another medical application, the ROI and background thresholds can be MM intensities set to optimize, for example, an MRI 3D visualization of a joint or of at least a portion of a brain. In other embodiments, to, for example, distinguish between multiple tissue types, two or more (e.g., multiple) ROI thresholds can be defined and used together with the background threshold in segmentation, rendering a 3D image and/or generating a 3D model.

In an image processing step 210, the processor utilizes the segmentation and/or the thresholds to generate a processed 3D image. The processing may include reducing the feature values of voxels classified as background voxels having feature values which satisfy the ROI feature threshold value but do not satisfy the background feature threshold value. Thus, a 3D model generated based on the processed image (e.g., image with altered feature values) and the ROI feature threshold may visualize tissue features inside ROIs (e.g., bone tissue) when irrelevant or undesired background information is suppressed.

The processed 3D image of image processing step 210 may be used by the processor for computing a 3D model, at a 3D model creation step 212. At 3D model creation step 212, the processor, in some embodiments, runs a 3D model generation application (e.g., a surface building algorithm), such as a marching cubes algorithm described by Lorensen E. and Harvey E. in a paper titled, "Marching cubes: A high resolution 3D surface construction algorithm," published in ACM SIGGRAPH Computer Graphics, Volume 21 (4): 163-169, July 1987, the content of which is incorporated herein by reference. As described above, the marching cubes

algorithm may use a threshold value, e.g., the lowest TH ROI value, to generate the 3D model. In some embodiments, the 3D model comprises a surface of an anatomy in the ROI (e.g., spine) and possibly comprising artificial elements as well, depending on TH ROI and TH background threshold settings. An example of a generated 3D model is shown in FIGS. 3A-3C, described below.

3D model cropping step 222 and 3D model saving step 224 include generating a suitable file of the 3D model, to be used, for example, with a near eye display (e.g., of head-mounted display unit 28) in image-guided surgery and/or with any other display such as a work station display. In an optional 3D model cropping step 222, the processor may define or may receive a definition (e.g., via user input) of a cropping box, such as shown in FIG. 4, and may crop the 3D model accordingly. Finally, the processor saves the 3D model (e.g., cropped 3D model) into a file of a suitable format (e.g., PLY format), in a memory, to be later uploaded to the processor or any other processor (e.g., a dedicated processor of a near-eye display of a head-mounted display unit 28).

In optional 3D image rendering step 208, the 3D image obtained medical image obtaining in step 202 is displayed on a monitor as a 3D visualization (e.g., a 3D volume rendering). A user, such as a medical professional can optimize the 3D visualization for the required or desired medical application (e.g., for spine image-guided surgery) by adjusting the two feature thresholds obtained (e.g., received, uploaded, etc.) in thresholds obtaining step 206. According to some embodiments, the initial 3D rendering is performed based on predetermined default values for the TH1 ROI and TH2 Background. According to some embodiments, the thresholds may be iteratively adjusted by the user. Final threshold values may be set or determined, for example, once a user input indicating that the threshold adjustment process is complete or a user request for generating of the 3D model is received.

In optional image cropping step 218, the processed 3D image obtained in image processing step 210 is displayed on a monitor so that the user can crop the portion of processed 3D image inputted for the modeling step (e.g., instead of the processor performing this step as part of, for example, image processing step 210). According to some embodiments, the cropping may be performed following optional 3D image rendering step 208 or in the frame of optional 3D image rendering step 208 and/or prior to image processing step 210.

In an additional optional step (not illustrated), the generated 3D model is displayed on a display (e.g., a stationary display such as a workstation and/or a near-eye display of head-mounted display unit 28) for the user's review.

The flowchart of FIG. 2 is presented by way of example and is simplified for clarity of presentation. The process may include additional or alternative steps, such as generating a 3D model from another imaging modality (e.g., based on a volume rendering obtained using ultrasound or MRI or fluoroscopy), or directly from high resolution segmentation, skipping the step of volume rendering, as described above.

FIGS. 3A-3C are exemplary Graphical User Interface (GUI) displays showing 3D renderings based on processed CT images (e.g., segmented CT images) of a spine that can be used for selecting or determining optimal or desired ROI intensity threshold and background intensity threshold for computing the 3D model, in accordance with embodiments of the disclosure. The 3D renderings may be viewed by a user that can determine or adjust the two thresholds of the thresholds obtaining step 206 above. The user may set or

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adjust the ROI and background intensity thresholds by adjusting a GUI element, such as sliding ROI TH sliders (302, 312, 322) and background TH sliders (304, 314, 324), respectively, at the bottom of the 3D rendering screen, respectively. The 3D rendering will change in correspondence to the change in thresholds and based on the segmentation, visualizing the data that will be included in the 3D model for each selection of thresholds or for each selected combination of thresholds. The user may then optimize the visualization or rendering while the thresholds values for the optimized or desired rendering will be provided as input to the 3D image processing and/or modeling. In FIGS. 3A-3C, the value of a threshold increases as the threshold slider is slid to the left. The slider may be moved by touching and dragging on the screen and/or by use of a computer mouse, keyboard, joystick, voice-activated control, or other input device.

FIG. 3A is a graphic display of a rendering 310 of a 3D image with threshold sliders set as follows: ROI TH slider 302 at 100%, meaning ROI intensity TH value=minimum scan radiodensity value and all is shown within the segmented area, and background threshold slider 304 at 0%, meaning background intensity TH value=maximum scan radiodensity value and all information outside the segmented portion is blocked.

FIG. 3B is a graphic display of a rendering of a 3D image with threshold sliders set as follows: ROI threshold slider 312 is set at 100%, meaning ROI intensity TH value=minimum scan radiodensity value and all tissue is shown within segmented portion. Background threshold slider is set at 22%, meaning background intensity TH value is high but such that metal retractors 318 and clamp 320 outside the segmented spine are shown.

FIG. 3C is a graphic display of a rendering of a 3D image with threshold sliders set as follows: ROI threshold slider is set at 92%, meaning ROI intensity TH value=low scan radiodensity value and most tissue is shown within segmented portion. Background threshold slider is set at 47%, meaning background intensity TH value is of mid value but such that the heads of metal screws 340 implanted in the bone protruding from the spine are shown.

Due to the relatively large background threshold, e.g., that is set at about the middle of the intensity range, irrelevant or undesired signals in the background, such as image noise and soft and bone tissue are not shown, making the disclosed processed 3D image particularly well visualizing the ROIs only.

In some embodiments, in generating the rendering, the one or more ROI feature thresholds would be applied to the one or more ROIs respectively while the background feature threshold would be applied only to the background portion of the image.

In some embodiments, as illustrated in FIGS. 3A-3C, when generating the rendering, the one or more ROI feature thresholds would be applied to the one or more ROIs respectively, while the background feature threshold would be applied to the entire image (i.e., including the one or more ROIs). In such a case, in the one or more ROIs, rendering may be performed based on the lowest threshold applied (i.e., the lowest of the respective ROI feature threshold and the background feature threshold). In such embodiment, the user may view a rendering or visualization of the medical image without the effect of the segmentation by simply setting the background feature threshold to a value lower than or equal to the one or more ROI feature thresholds. Following that, a single feature threshold value would apply

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to the entire scan or image without a distinction between different portions of the image (e.g., ROI vs. background).

FIG. 4 is a graphical user interface display showing a 3D model 400 of a volume of interest, the 3D model 400 computed based on processed 3D image of FIG. 3C, in accordance with an embodiment of the disclosure. As seen, the 3D model 400 is located inside a box 405 having marked facets 415. The disclosed 3D model clearly presents a spine section implanted with metal screws 440 with clean background (e.g., empty of, or without, irrelevant or undesired information). Such 3D model is suitable for presentation and/or augmentation with the optically viewed (e.g., with the near-eye display device) anatomical portion of the patient.

FIG. 5 is a flowchart of steps performed to generate a three-dimensional (3D) model based on a 3D image of a portion of a body of a patient using selected threshold values, in accordance with an embodiment of the disclosure. The method according to the presented embodiment carries out a process that begins with obtaining a 3D medical image having feature values, of at least a portion of an organ or anatomy of a patient (e.g., an entire spine or region of a spine, such as a lumbosacral region, a cervical region, a thoracic region, and/or a sacroiliac region), at a 3D medical image receiving step 502. The features may be directly extracted from the medical image, as in the case the feature is intensity, or may require an additional operation of processing or computation, for example, as in the case the feature is gradients. The 3D medical image may be obtained, for example, via provided input, uploaded or downloaded via the internet. In some embodiments the 3D image may be obtained by capturing the medical image via a medical imaging device.

Next, a processor segments the 3D medical image to define one or more ROIs (e.g., portions of spine regions), at 3D medical image segmentation step 504.

At a ROI feature threshold value determination step 506, the processor determines at least one ROI features threshold, such as image intensity threshold value. At a background feature threshold value determination step 508, the processor determines a background features threshold, such as image background intensity threshold value.

The determination of the at least one ROI feature threshold and of the background feature threshold of ROI feature threshold value determination step 506 and background feature threshold value determination step 508 may include receiving input values (e.g., from the user) for the at least one ROI feature threshold or for the background feature threshold or for both. In some implementations, the receiving of the input values from the user comprises generating a GUI element to be displayed to the user, such as ROI TH sliders 302, 312, 322 and background TH sliders 304, 314, 324 of FIGS. 3A-3C. The GUI element may allow the user to adjust the input values, while the 3D rendering and displaying of the 3D rendering to the user are iteratively performed in correspondence to the user adjustment of the input values. In some embodiments, the one or more ROI feature thresholds and the background feature threshold are predetermined. In some embodiments, a default value for some or for each of the one or more ROI feature thresholds and the background feature threshold is predetermined. A 3D rendering based on these default values may be displayed to the user as a default 3D rendering. The user may then adjust the thresholds values as described above.

At model feature threshold value selection step 510, the processor selects or determines the threshold to be input to a 3D model generation algorithm based on the determined

one or more ROI feature thresholds and the background feature threshold. In some embodiments, the lowest ROI feature threshold may be selected or determined as the input threshold for the 3D model generation. In some embodiments, the lowest of the one or more ROI feature thresholds and the background feature threshold may be selected.

At threshold values checking step **512**, the processor checks if the background feature threshold value is greater than the threshold value selected to be provided to the 3D model generation algorithm (model feature threshold value selection step **510** above).

If the result of the check performed in threshold values checking step **512** is positive, i.e., the background feature threshold value is greater than the threshold value selected in model feature threshold value selection step **510**, then selected feature values in the background (e.g., feature values associated with voxels segmented as background (3D) medical image segmentation step **504** above)) are reduced or suppressed in a step **514**. Background feature values which satisfy the model generation threshold (i.e., the threshold selected to be input to the 3D model generation algorithm in model feature threshold value selection step **510** above) but do not satisfy the background feature threshold are reduced to a value equal to or below the model generation threshold. In some embodiments, such background feature values are reduced to a value equal to the value of the model generation threshold minus a constant. Thus, background elements (e.g., voxels) having feature values which do not satisfy the background feature threshold, are not considered for the model generation, or excluded from being included in the model generation algorithm input.

At a step **516**, a 3D model is generated by a model generation algorithm that accepts a single 3D image feature threshold value selected at model feature threshold value selection step **510**. An example of such a model is the aforementioned marching cubes algorithm.

Finally, the processor displays (or generates an output for display) the generated 3D model to the user, at a 3D model displaying step **518**.

The flowchart of FIG. **5** is presented by way of example and is simplified for clarity of presentation. The process may include alternative steps, such as determining threshold values that are from different medical modalities.

FIGS. **6A-6D** are example GUI displays **600** that include various views of a patient's spine, in accordance with an embodiment of the disclosure. FIGS. **6A-6D** show an illustration of image data displayed during a medical procedure for inserting pedicle screws into a patient's spine. The image data include a display of a 3D model or rendering **601** of an ROI (e.g., a portion of the patient's spine), and virtual images of a tool **602** and a pedicle screw implant **604**. In some embodiments, 3D model **601** may be generated as disclosed hereinabove. The GUI displays **600** in FIGS. **6A-6D** further include various views, represented as images, of the patient spine and pedicle screw implant **604**: X-ray lateral view **606**, X-ray anteroposterior (AP) view **608**, 3D lateral view **610** and 3D AP view **612**, displayed in different combinations.

The displays shown in FIGS. **6A-6D** are not augmented-reality displays and may be displayed, for example, on a workstation. However, these displays may be displayed in a see-through display of a wearable, hands-free and/or head-mounted display system while 3D model **601** is overlaid on the patient anatomy, optionally in-alignment, and as described hereinabove. The different views (**606**, **608**, **610** and/or **612**) may be displayed on a different section of the

see-through display, e.g., in a top portion or a side portion of the display and such that they will not interfere with the user's view (e.g., direct see-through view) of the physical ROI through the display.

A user may select which view will be presented in each view display window, such as windows **614** and **616**. The different views may be presented in the GUI display, for example, via a drop-down menu, such as drop-down menu **618**, associated with each view display window. The GUI display **600** shown in FIGS. **6A-6D** includes two view display windows. Other embodiments may include a different number of view display windows. In some embodiments, the user may select which view to display in which view display window (e.g., via a drop-down menu) and switch between the views during the procedure, as desired. Accordingly, the user may generate different combinations of views during the procedure, allowing him or her to receive maximal and proper information for each maneuver, occurrence and/or phase of the procedure.

X-ray views **606** and **608** are X-ray like images or virtual X-ray images derived from the 3D medical image as viewed from a fixed position, specifically lateral and AP respectively. Lateral and AP X-ray like or virtual X-ray views may facilitate navigation while AP view may allow, for example, symmetric screw insertion planning (e.g., during minimally invasive surgical (MIS) procedures). The virtual X-ray images may be generated by the processor from the 3D medical image by generating digitally reconstructing radiographs. In some embodiments, the processor may utilize the segmentation to digitally reconstruct radiographs from the segmented 3D medical image. The voxel feature values segmented as background (i.e., not ROI) may be set to zero, thus generating X-ray views of the ROI only (e.g., without noise or with reduced noise), as shown, for example, in FIGS. **6A-6D**. In some embodiments, a GUI element such as slider **620** may be generated to allow the user to adjust the DRR images visibility threshold (e.g., sum of voxels intensity values). Each change of visibility threshold may cause the processor to generate a new DRR based on the adjusted visibility threshold.

In some embodiments, X-ray views **606** and **608** may display 2D X-ray images of the patient anatomy captured by an X-ray imaging device such as a fluoroscope. In some embodiments, the X-ray images may be registered with the 3D medical image.

3D views **610** and **612** display the 3D model as viewed from a fixed position, specifically lateral and AP respectively. The 3D view may facilitate visualization and orientation and may simplify entry point location.

Although FIGS. **6A-6D** show X-ray views and 3D views from lateral and AP points of view, other points of view or angles of view may be displayed alternatively or additionally. It should be noted that lateral and AP points of view may be specifically advantageous for medical procedures or interventions which the professional may perform via a lateral approach (e.g., by positioning the patient on the side of his body), such as a Lateral Lumbar Interbody Fusion (LLIF) procedure.

In some embodiments, as indicated above, the user may select which views will be displayed in each window of the display, such as windows **614** and **616**. FIGS. **6A-6D** show such various combinations which may be advantageous to the user during a medical procedure or intervention (e.g., surgical or non-surgical therapeutic or diagnostic procedure). FIG. **6A** shows a combination of X-ray views **606** and **608** from different points of view, specifically lateral and AP. FIG. **6B** shows a combination of 3D views **610** and **612** from

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different points of view, specifically lateral and AP. FIG. 6C shows a combination of lateral views 606 and 610 via different types of view (e.g., imaging modalities or techniques), specifically X-ray and 3D, respectively. FIG. 6D shows a combination of AP views 608 and 612 via different types of view, specifically X-ray and 3D, respectively. It should be noted that 3D view may provide surface-related data of a bone structure, while X-ray view may provide depth-related data of the bone structure. Displaying a specific type of view, e.g., X-ray view or 3D view, from multiple points of view or a specific point of view, e.g., lateral or AP, via different types of view at the same time may be advantageous and may facilitate the procedure or intervention or even better the procedure or intervention outcome by providing the required information and/or more information and better visualization of the ROI to the professional performing the procedure or intervention.

While images of views 606, 608, 610 and 612 are static, virtual images of the tool 602 and screw implant 604 may be dynamically augmented on images of views 606, 608, 610 and 612 presenting the actual location and navigation of the tool and/or screw implant with respect to the patient anatomy. Such display may be facilitated via a tracking system and as described hereinabove. Since screw implant 604 is attached as a straight extension to tool 602 and since its dimensions may be predefined, there is no need for tracking of screw implant 604 and tracking of only tool 602 may suffice.

In some embodiments, the GUI may include one or more slice views of the 3D medical image. In some embodiments, a sagittal slice view and an axial slice view may be generated and displayed. A virtual image representing a tool and/or a virtual image representing an implant, such as a pedicle screw may be augmented on the slice views, while navigated by a professional during a medical procedure or intervention. In some embodiments, the displayed slice may be selected and/or generated based on or according to the position of the navigated tool. In some embodiments, the displayed slice may be selected and/or generated such that a predefined axis of the tool is aligned with the slice plane. For example, for an elongated tool, such as a screwdriver, the tool longitudinal axis may be predefined as the tool axis. Thus, for each tracked and/or recorded change of position of the tool a new slice may be generated and displayed.

In some embodiments, for each anatomical view only slices of the specific anatomical view would be generated and displayed in response to the manipulation of the tool. For example, for an axial view, only axial slices would be displayed according to the axial component of the tool current position. In some embodiments, in one or more anatomical views, slices may be generated based on tool movement in two or three anatomical planes. For example, in an axial view, for a current position of the tool having components in the axial and coronal planes, a slice positioned axially and coronally in accordance with the axial and coronal tool position, would be generated and displayed. As another example, in an axial view, for a current position of the tool having components in the axial, coronal and sagittal planes, a corresponding slice would be generated for the axial view. Thus, for example, for a tool positioned in a sagittal plane, the axial slice view and the sagittal slice view may be similar or even identical. Such anatomical slice view which allows slice generation in multiple planes and not just in the predefined view plane provides better visualization and may facilitate navigation.

In some embodiments, the display of the various views may be manipulated by a user. In some embodiments a view

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may be mirrored, e.g., with respect to an anatomical plane of the patient. For example, a mirroring command button may be included in the GUI. Mirroring of a lateral view may be performed, for example, with respect to a plane of the patient body. In some embodiments, the view may be rotated by a specific angle, e.g., by 180°, with respect to an anatomical axis, for example. Such rotating may be performed by pressing a dedicated command button generated as part of the GUI, for example. Specifically, such rotating of an AP view may provide a view rotated by 180° with respect to the longitudinal anatomical axis (i.e., head to toe). Such manipulation may be advantageous, for example, when the medical image and its derived images are displayed from a side opposite to the desired side (e.g., left side or right side of the patient or upper side or lower side of the patient). In some embodiments, the various views may be panned and zoomed in and out, e.g., via dedicated command buttons generated as part of the GUI.

While examples of the disclosed technique are given for body portion containing spine vertebrae, the principles of the system, method, and/or disclosure may also be applied to other bones and/or body portions than spine, including hip bones, pelvic bones, leg bones, arm bones, ankle bones, foot bones, shoulder bones, cranial bones, oral and maxillofacial bones, sacroiliac joints, etc.

The disclosed technique is presented with relation to image-guided surgery systems or methods, in general, and accordingly, the disclosed technique of visualization of medical images should not be considered limited only to augmented reality systems and/or head-mounted systems. For example, the technique is applicable to the processing of images from different imaging modalities, as described above, for use in diagnostics.

The terms “top,” “bottom,” “first,” “second,” “upper,” “lower,” “height,” “width,” “length,” “end,” “side,” “horizontal,” “vertical,” and similar terms may be used herein; it should be understood that these terms have reference only to the structures shown in the figures and are utilized only to facilitate describing embodiments of the disclosure. Various embodiments of the disclosure have been presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the disclosure. The ranges disclosed herein encompass any and all overlap, sub-ranges, and combinations thereof, as well as individual numerical values within that range. For example, description of a range such as from about 5 to about 30 degrees should be considered to have specifically disclosed subranges such as from 5 to 10 degrees, from 10 to 20 degrees, from 5 to 25 degrees, from 15 to 30 degrees etc., as well as individual numbers within that range (for example, 5, 10, 15, 20, 25, 12, 15.5 and any whole and partial increments therebetween). Language such as “up to,” “at least,” “greater than,” “less than,” “between,” and the like includes the number recited. Numbers preceded by a term such as “about” or “approximately” include the recited numbers. For example, “approximately 2 mm” includes “2 mm.” The terms “approximately,” “about,” and “substantially” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result.

In some embodiments, the system comprises various features that are present as single features (as opposed to multiple features). For example, in one embodiment, the system includes a single HMD, a single camera, a single processor, a single display, a single fiducial marker, a single

imaging device, etc. Multiple features or components are provided in alternate embodiments.

In some embodiments, the system comprises one or more of the following: means for imaging (e.g., a camera or fluoroscope or MRI machine or CT machine), means for calibration or registration (e.g., adapters, markers, objects), means for fastening (e.g., anchors, adhesives, clamps, pins), etc.

The processors described herein may include one or more central processing units (CPUs) or processors or microprocessors. The processors may be communicatively coupled to one or more memory units, such as random-access memory (RAM) for temporary storage of information, one or more read only memory (ROM) for permanent storage of information, and one or more mass storage devices, such as a hard drive, diskette, solid state drive, or optical media storage device. The processors (or memory units communicatively coupled thereto) may include modules comprising program instructions or algorithm steps configured for execution by the processors to perform any of all of the processes or algorithms discussed herein. The processors may be communicatively coupled to external devices (e.g., display devices, data storage devices, databases, servers, etc. over a network via a network communications interface.

In general, the algorithms or processes described herein can be implemented by logic embodied in hardware or firmware, or by a collection of software instructions, possibly having entry and exit points, written in a programming language, such as, for example, Python, Java, Lua, C, C #, or C++. A software module or product may be compiled and linked into an executable program, installed in a dynamic link library, or may be written in an interpreted programming language such as, for example, BASIC, Perl, or Python. It will be appreciated that software modules may be callable from other modules or from themselves, and/or may be invoked in response to detected events or interrupts. Software modules configured for execution on computing devices may be provided on a computer readable medium, such as a compact disc, digital video disc, flash drive, or any other tangible medium. Such software code may be stored, partially or fully, on a memory device of the executing computing device, such as the processing system 50, for execution by the computing device. Software instructions may be embedded in firmware, such as an EPROM. It will be further appreciated that hardware modules may be comprised of connected logic units, such as gates and flip-flops, and/or may be comprised of programmable units, such as programmable gate arrays or processors. The modules described herein are preferably implemented as software modules but may be represented in hardware or firmware. Generally, any modules or programs or flowcharts described herein may refer to logical modules that may be combined with other modules or divided into sub-modules despite their physical organization or storage.

The various features and processes described above may be used independently of one another or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of this disclosure. In addition, certain method or process blocks or steps may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks, steps, or states relating thereto can be performed in other sequences that are appropriate. For example, described blocks, steps, or states may be performed in an order other than that specifically disclosed, or multiple blocks or states may be combined in a single block or state. The example blocks, steps, or states

may be performed in serial, in parallel, or in some other manner. Blocks, steps, or states may be added to or removed from the disclosed example embodiments. The example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed example embodiments.

Any process descriptions, elements, or blocks in the flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process.

It will be appreciated that the systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible or required for the desirable attributes disclosed herein. The various features and processes described above may be used independently of one another or may be combined in various ways. The section headings used herein are merely provided to enhance readability and are not intended to limit the scope of the embodiments disclosed in a particular section to the features or elements disclosed in that section.

Certain features that are described in this specification in the context of separate embodiments also may be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment also may be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination. No single feature or group of features is necessary or indispensable to each and every embodiment.

Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. In addition, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. In addition, the articles “a,” “an,” and “the” as used in this application and the appended claims are to be construed to mean “one or more” or “at least one” unless specified otherwise.

What is claimed is:

1. A computer-implemented method for improving display of 3D models in connection with image-guided surgery, the method comprising:

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obtaining a three-dimensional (3D) computed tomography image of a region of a spine of a patient, the 3D computed tomography image having intensity values; segmenting the 3D computed tomography image to define multiple regions of interest (ROIs) of the spine and a background region, wherein the background region comprises a portion of the 3D computed tomography image which is not an ROI;

determining, for each of the multiple ROIs of the spine, a ROI intensity threshold value that can be used as an input to control what portions of the ROI are included in a 3D rendering of the 3D computed tomography image;

determining a background intensity threshold value that can be used as an input to control what portions of the background region are included in the 3D rendering of the 3D computed tomography image;

generating the 3D rendering of the 3D computed tomography image, wherein the generation of the 3D rendering comprises:

- in each of the multiple ROIs of the spine of the 3D computed tomography image, rendering based on intensity values of the ROI that satisfy a lowest threshold value of the determined ROI intensity threshold value and the determined background intensity threshold value; and
- in the background region of the 3D computed tomography image, rendering based on intensity values of the background region that satisfy the determined background intensity threshold value; and

causing the 3D rendering to be output for display to a user, wherein the intensity threshold values are voxel values.

2. The computer-implemented method of claim 1, further comprising displaying the 3D rendering in alignment by performing registration of the 3D rendering with the region of the spine.

3. The computer-implemented method of claim 1, wherein the 3D rendering is a 3D model.

4. The computer-implemented method of claim 1, wherein the generating the 3D rendering comprises changing at least some of the intensity values of the 3D computed tomography image into a value which does not satisfy the lowest threshold value.

5. The computer-implemented method of claim 1, wherein the background region of the 3D computed tomography image comprises soft tissue.

6. The computer-implemented method of claim 1, further comprising:

- repeatedly adjusting one or more of the ROI intensity threshold values and the background intensity threshold value according to input from a user; and
- repeatedly generating the 3D rendering of the 3D computed tomography image based on the adjusted values.

7. The computer-implemented method of claim 1, wherein the multiple ROIs of the spine comprise various regions of the spine.

8. The computer-implemented method of claim 1, wherein the multiple ROIs of the spine comprise individual vertebrae of the spine.

9. The computer-implemented method of claim 1, further comprising:

- adjusting one or more of the ROI intensity threshold values and the background intensity threshold value according to input from a user;
- regenerating the 3D rendering of the 3D computed tomography image based on the adjusted values; and
- generating a 3D model based on the adjusted values.

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10. A computer-implemented method for improving display of 3D models in connection with image-guided surgery, the method comprising:

- obtaining a three-dimensional (3D) image of a region of a spine of a patient, the 3D image having intensity values,
- segmenting the 3D image to define multiple regions of interest (ROIs) of the spine and a background region, wherein the background region comprises a portion of the 3D image which is not an ROI;
- determining, for each of the multiple ROIs of the spine, a ROI intensity threshold value that can be used as an input to control what portions of the ROI are included in a 3D rendering of the 3D image;
- determining a background intensity threshold value that can be used as an input to control what portions of the background region are included in the 3D rendering of the 3D image;
- generating the 3D rendering of the 3D image, wherein the generation of the 3D rendering comprises:
 - in each of the multiple ROIs of the spine of the 3D image, rendering based on intensity values of ROI that satisfy a lowest threshold value of the determined ROI intensity threshold value and the determined background intensity threshold value; and
 - in the background region of the 3D image, rendering based on intensity values of the background region that satisfy the determined background intensity threshold value; and
- causing the 3D rendering to be output for display to a user.

11. The computer-implemented method of claim 10, wherein the intensity values are voxel values of the 3D image.

12. The computer-implemented method of claim 10, wherein the 3D image is a computed tomography image or a magnetic resonance image.

13. The computer-implemented method of claim 10, further comprising displaying the 3D rendering in alignment by performing registration of the 3D rendering with the region of the spine.

14. The computer-implemented method of claim 10, wherein the 3D rendering is a 3D model.

15. The computer-implemented method of claim 10, wherein the generating the 3D rendering comprises changing at least some of the intensity values of the 3D image into a value which does not satisfy the lowest threshold value.

16. The computer-implemented method of claim 10, wherein the background region of the 3D image comprises soft tissue.

17. The computer-implemented method of claim 10, further comprising:

- repeatedly adjusting one or more of the ROI intensity threshold values and the background intensity threshold value according to input from a user; and
- repeatedly generating the 3D rendering of the 3D image based on the adjusted values.

18. The computer-implemented method of claim 10, further comprising:

- adjusting one or more of the ROI intensity threshold values and the background intensity threshold value according to input from a user;
- regenerating the 3D rendering of the 3D image based on the adjusted values; and
- generating a 3D model based on the adjusted values.

19. The computer-implemented method of claim 10, wherein the multiple ROIs of the spine comprise various regions of the spine.

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20. The computer-implemented method of claim **10**, wherein the multiple ROIs of the spine comprise individual vertebrae of the spine.

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